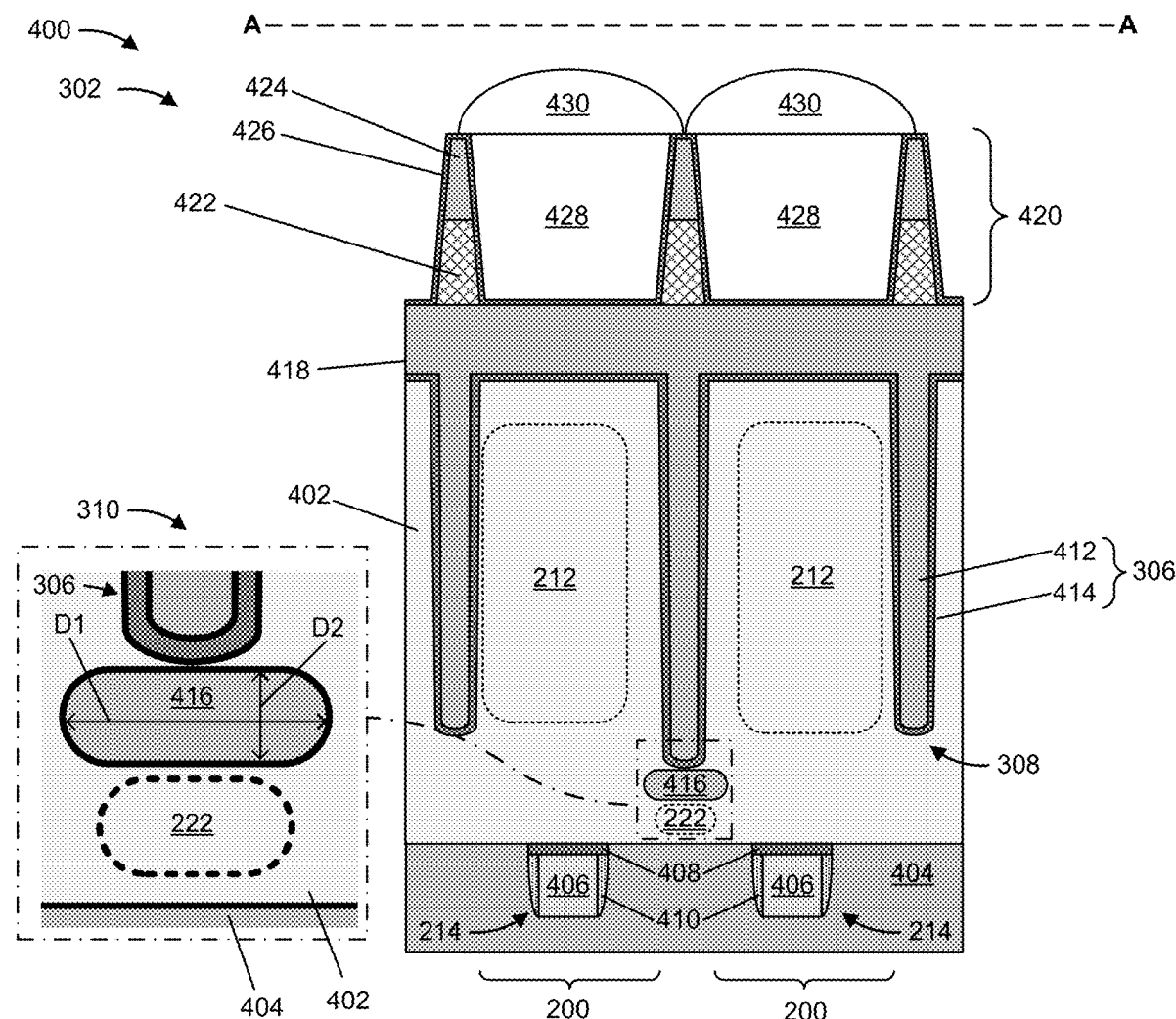


(43) **Pub. Date:** **Aug. 28, 2025**

Dielectric regions are formed in a semiconductor substrate of a semiconductor device. The dielectric regions may be formed in locations above which cross-road portions of a backside deep trench isolation (BDTI) structure are to be formed for a pixel sensor array of the semiconductor device. The dielectric regions may function as etch stop structures in the cross-road portions when etching the semiconductor substrate to form the trenches in which the BDTI structure is to be formed. In particular, the dielectric regions minimize and/or prevent the increase in trench depth of the trenches in the cross-road portions relative to the trench depth of the trenches in the non-cross-road portions.



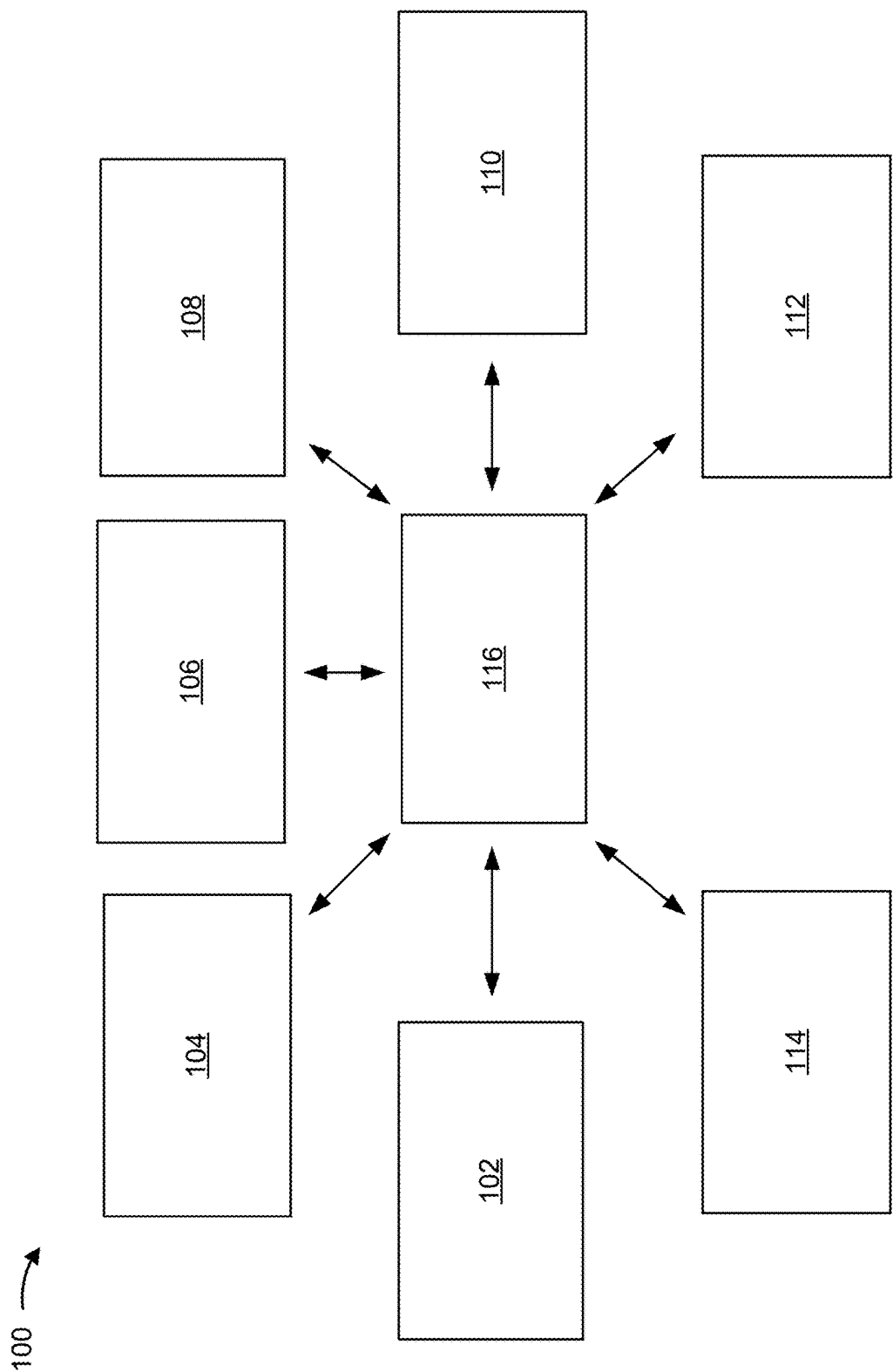


FIG. 1

200 →

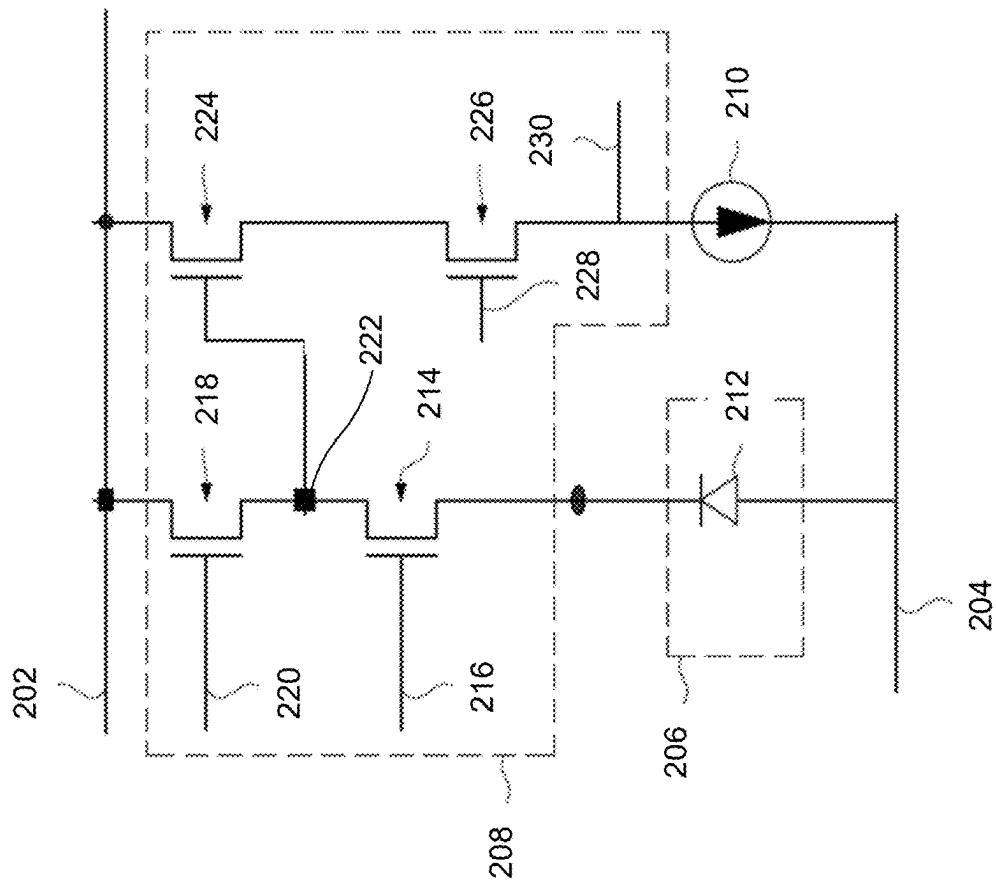


FIG. 2

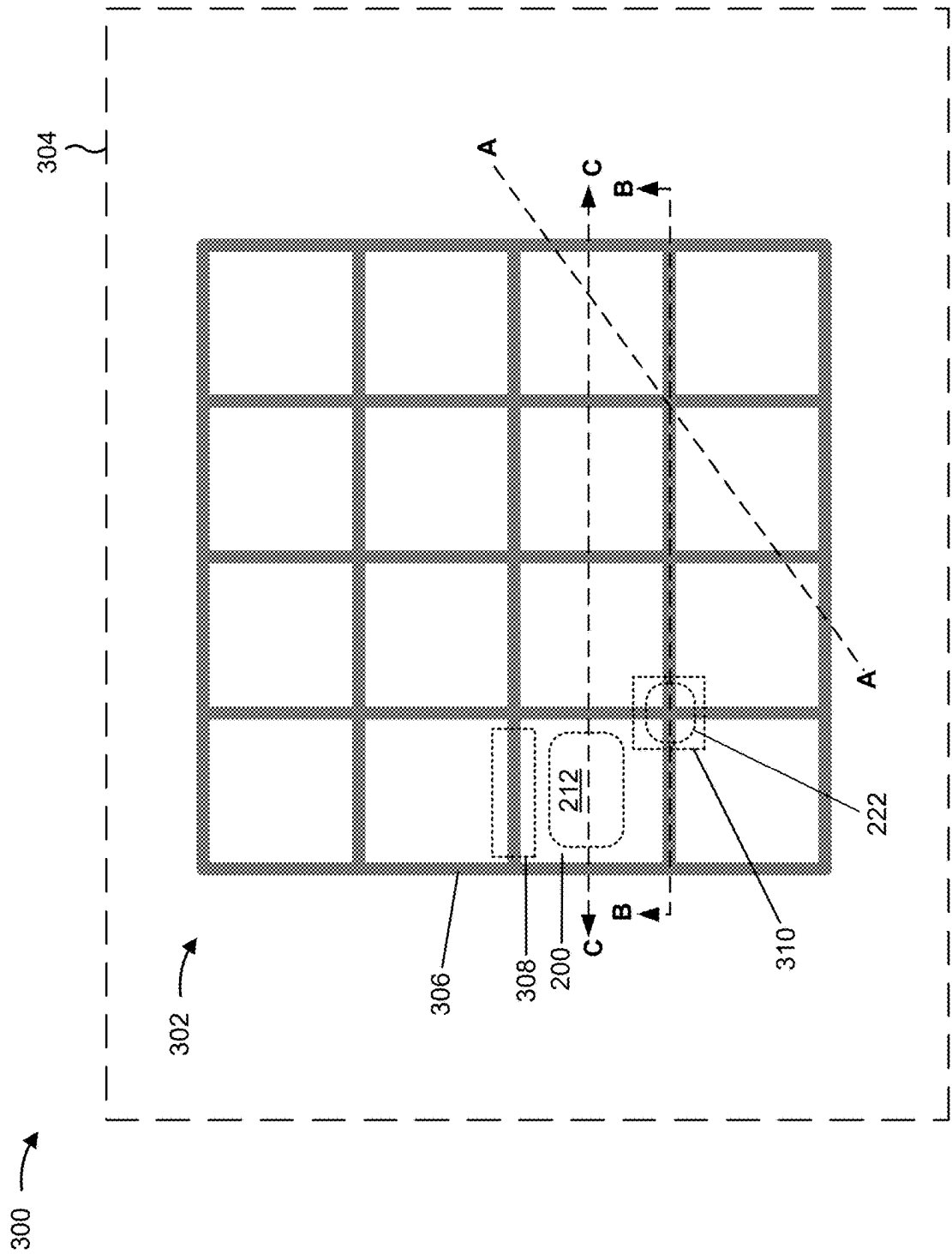


FIG. 3A

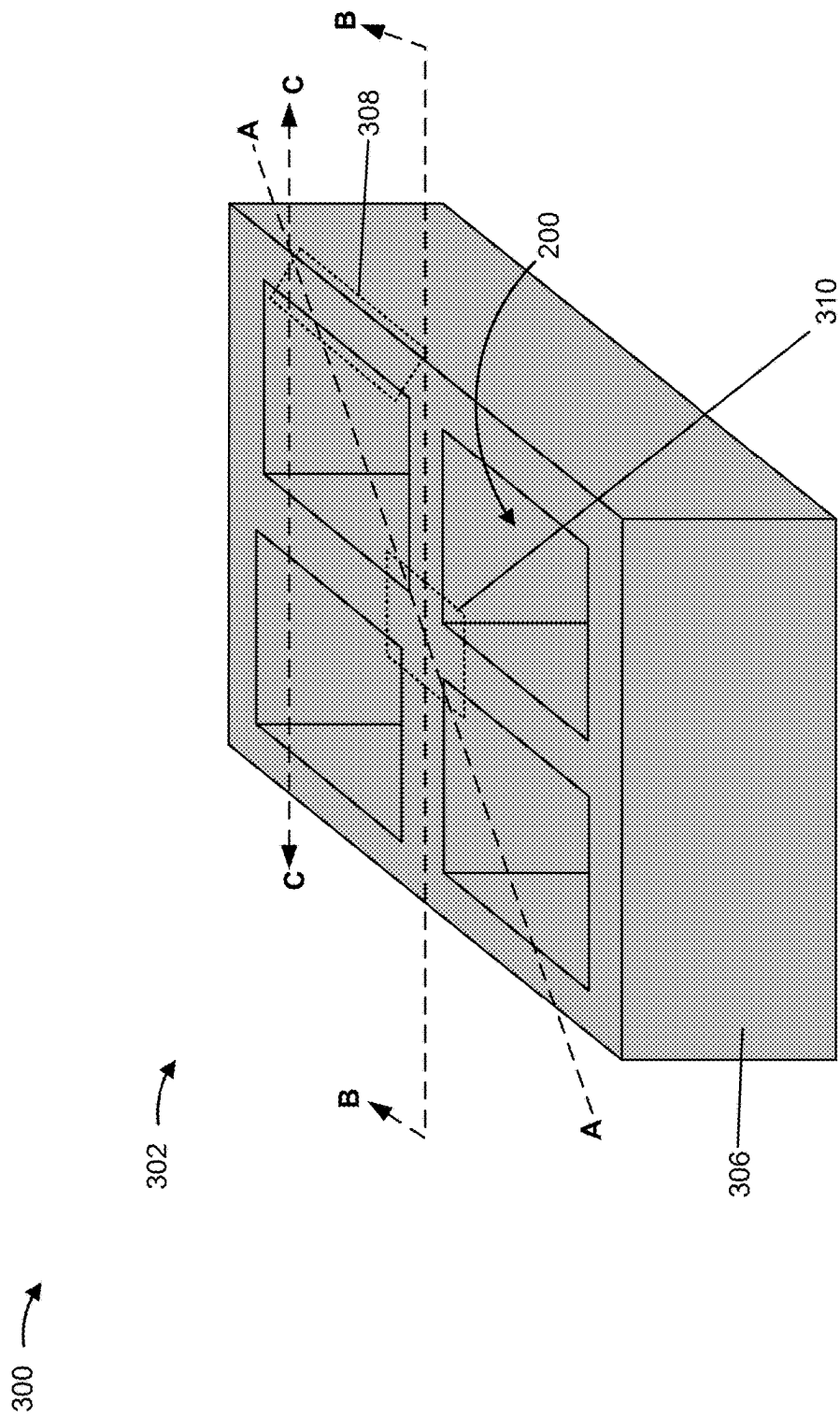


FIG. 3B

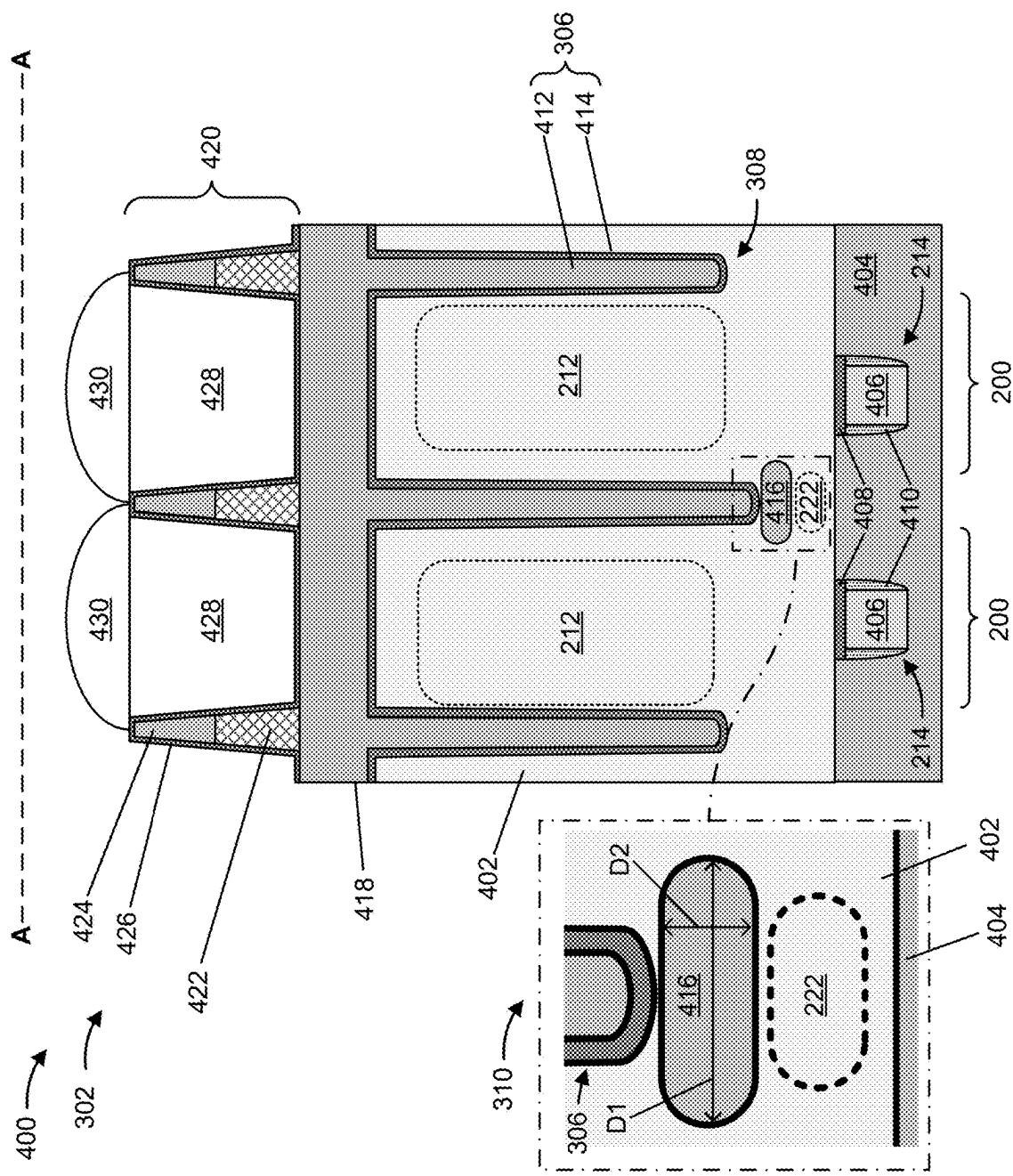


FIG. 4A

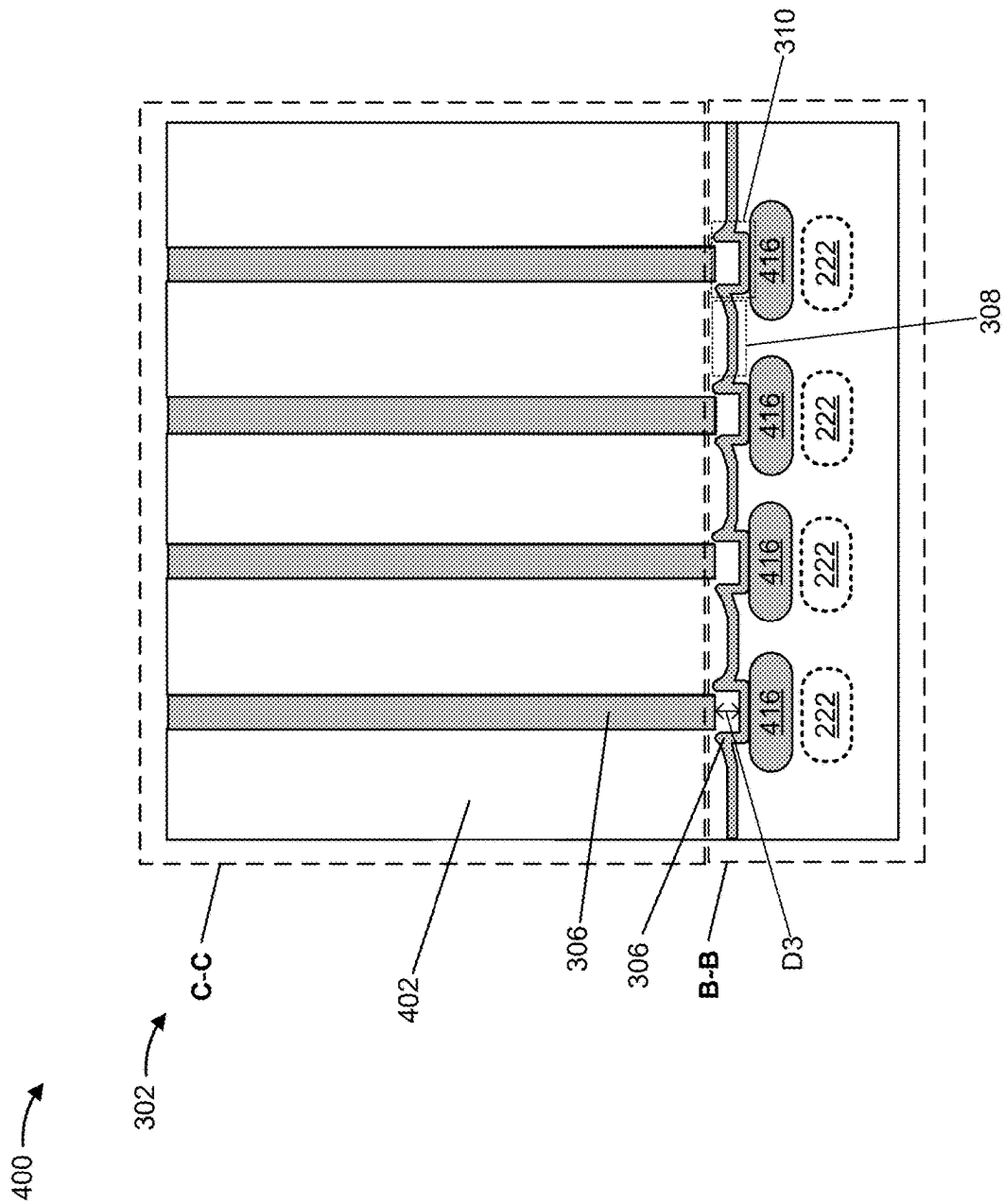
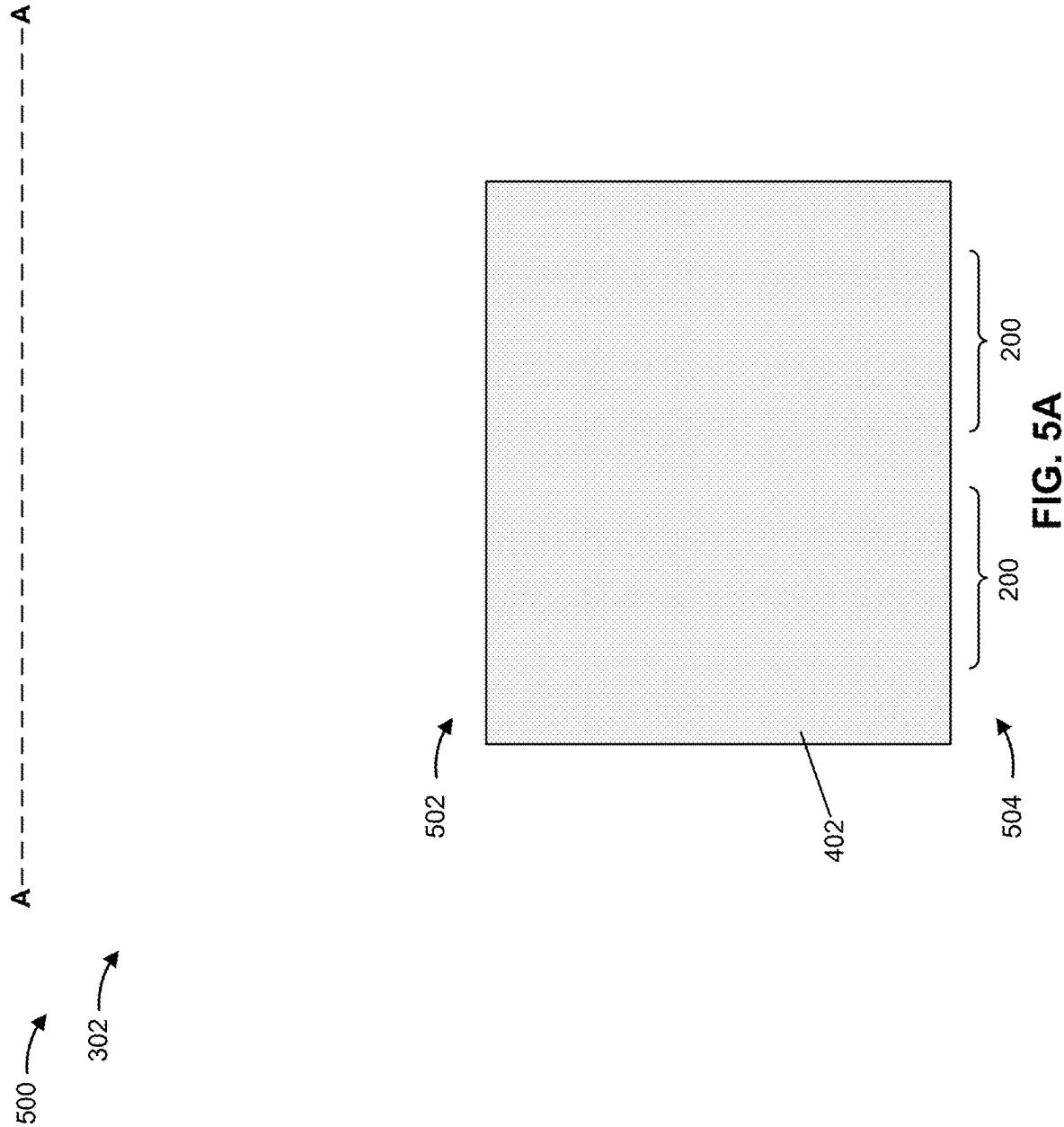
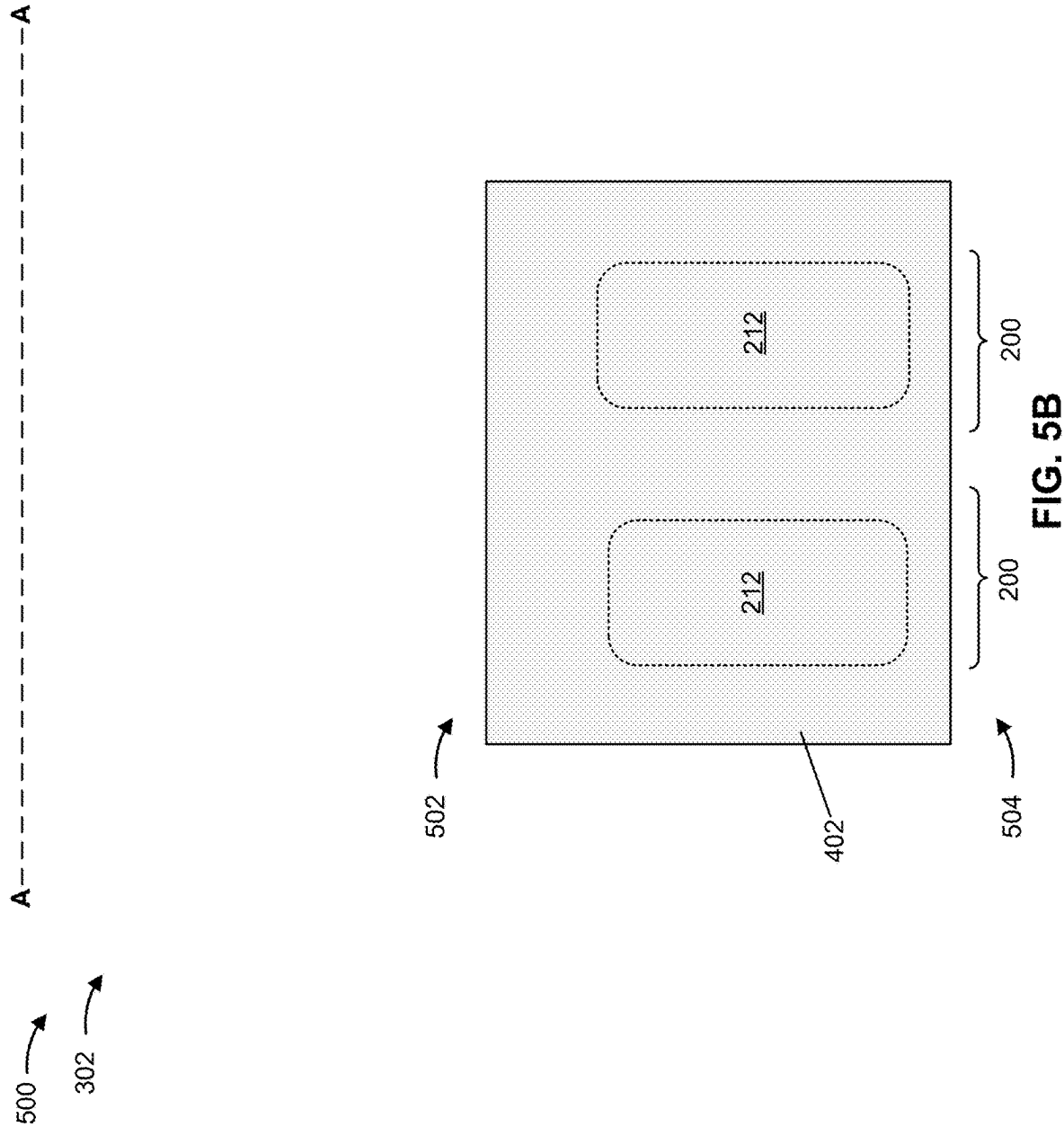
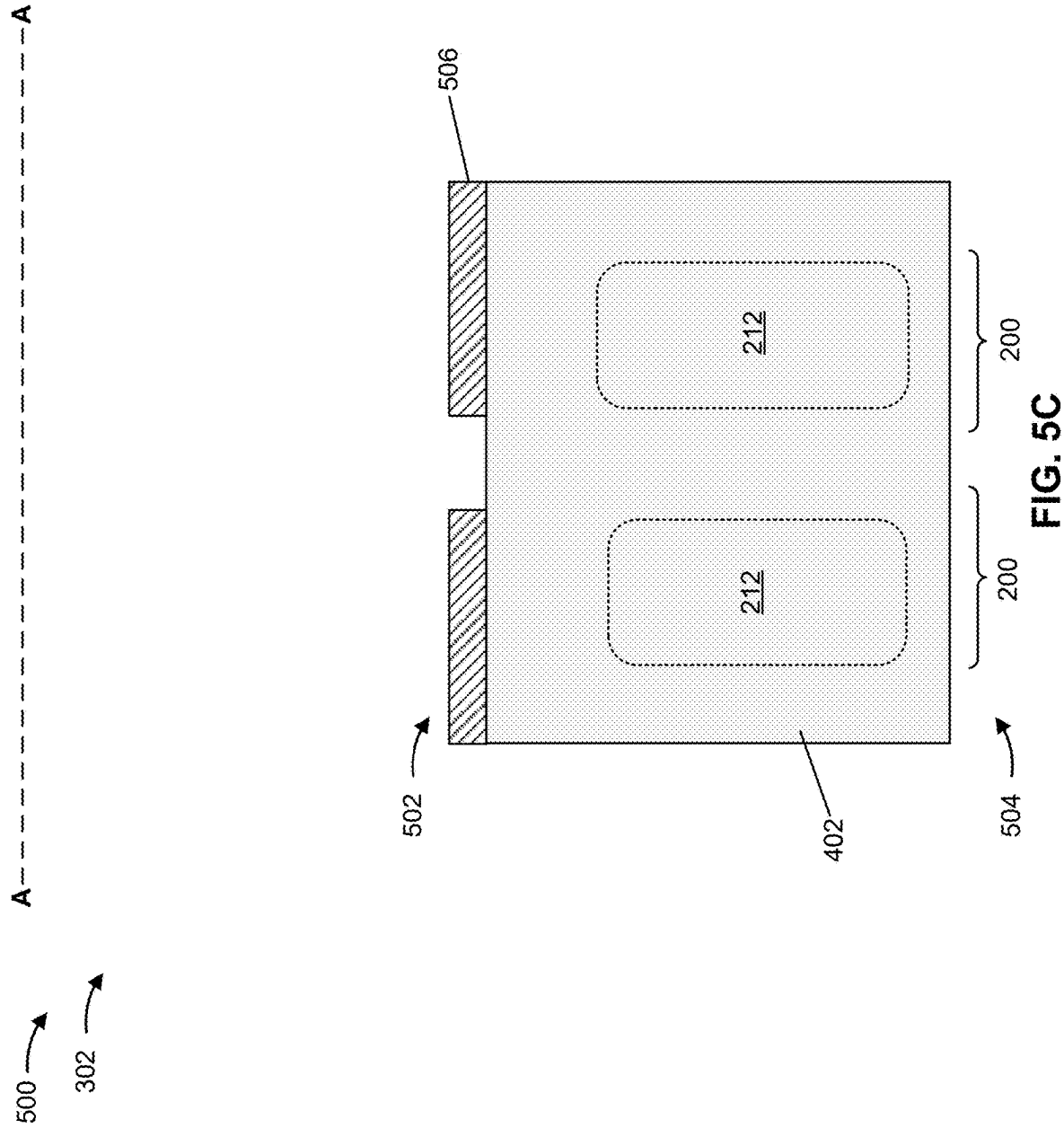
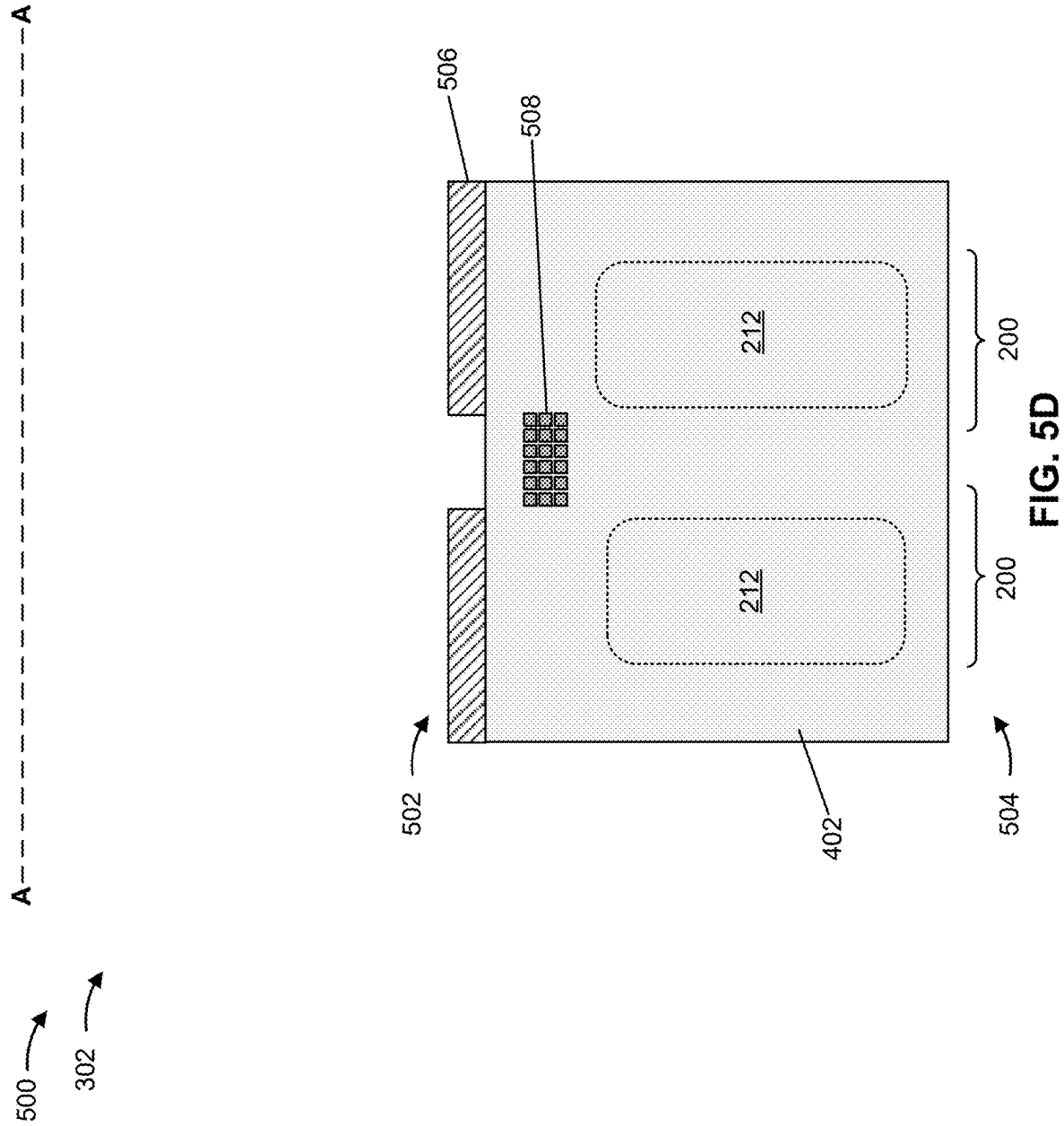


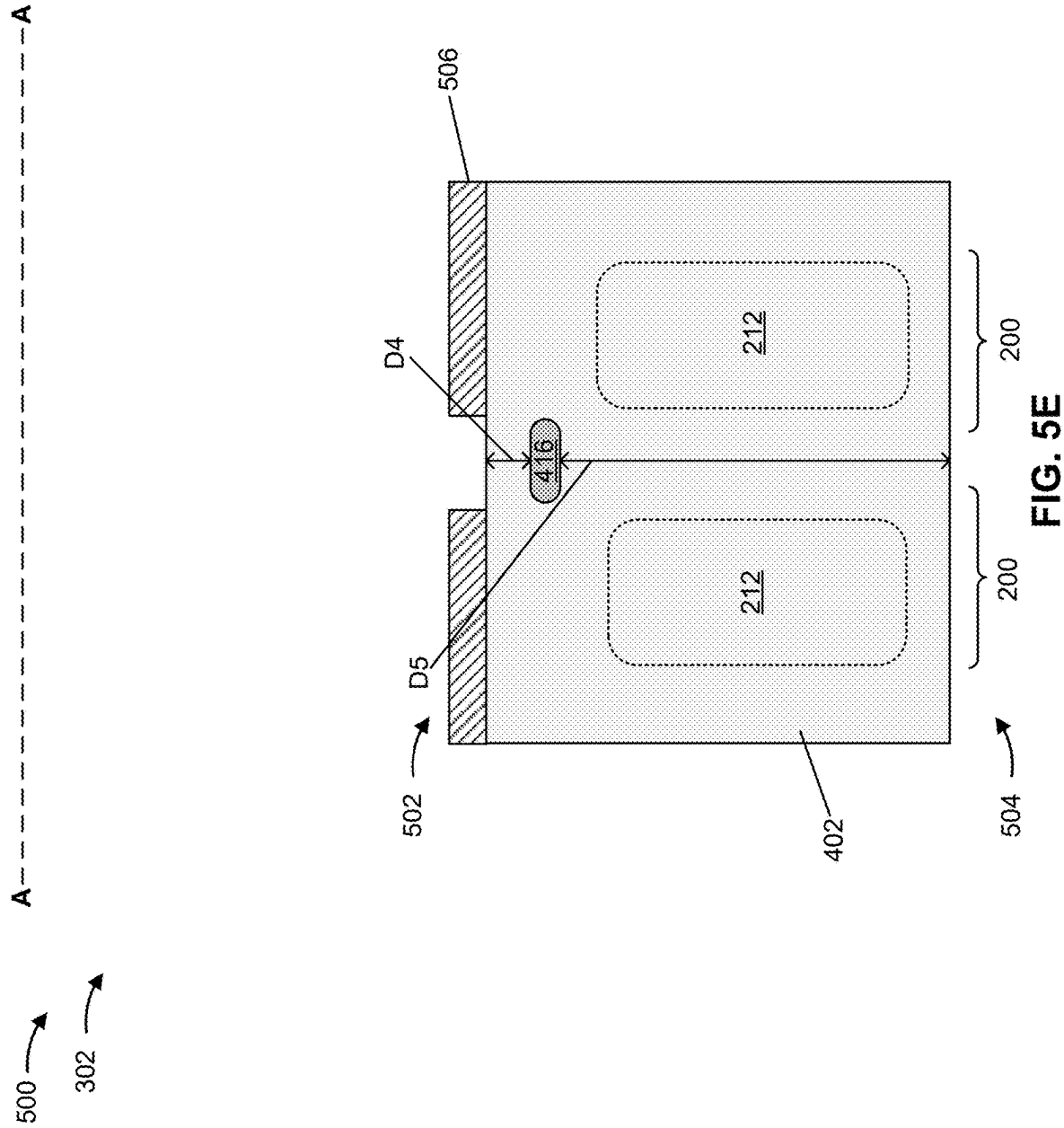
FIG. 4B











A-----A

500
302

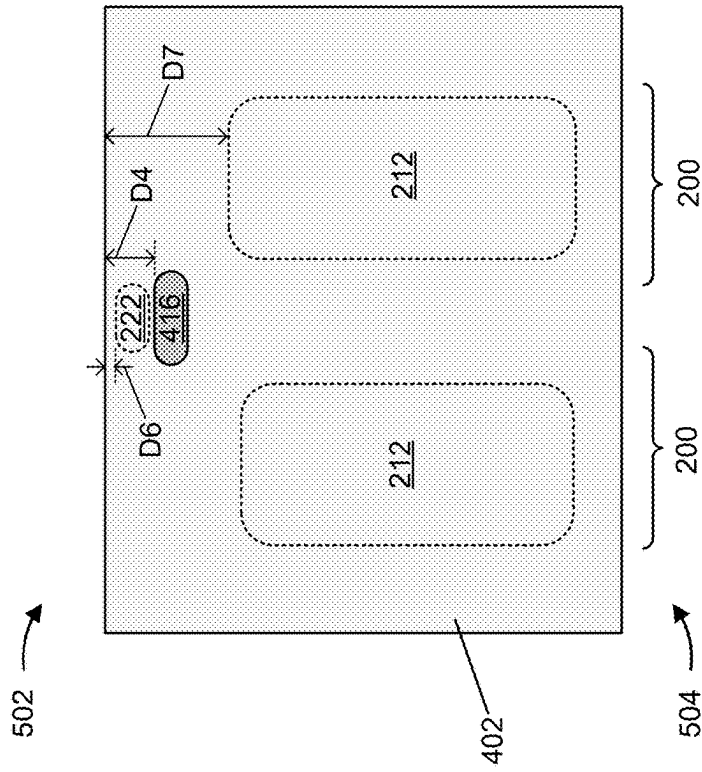
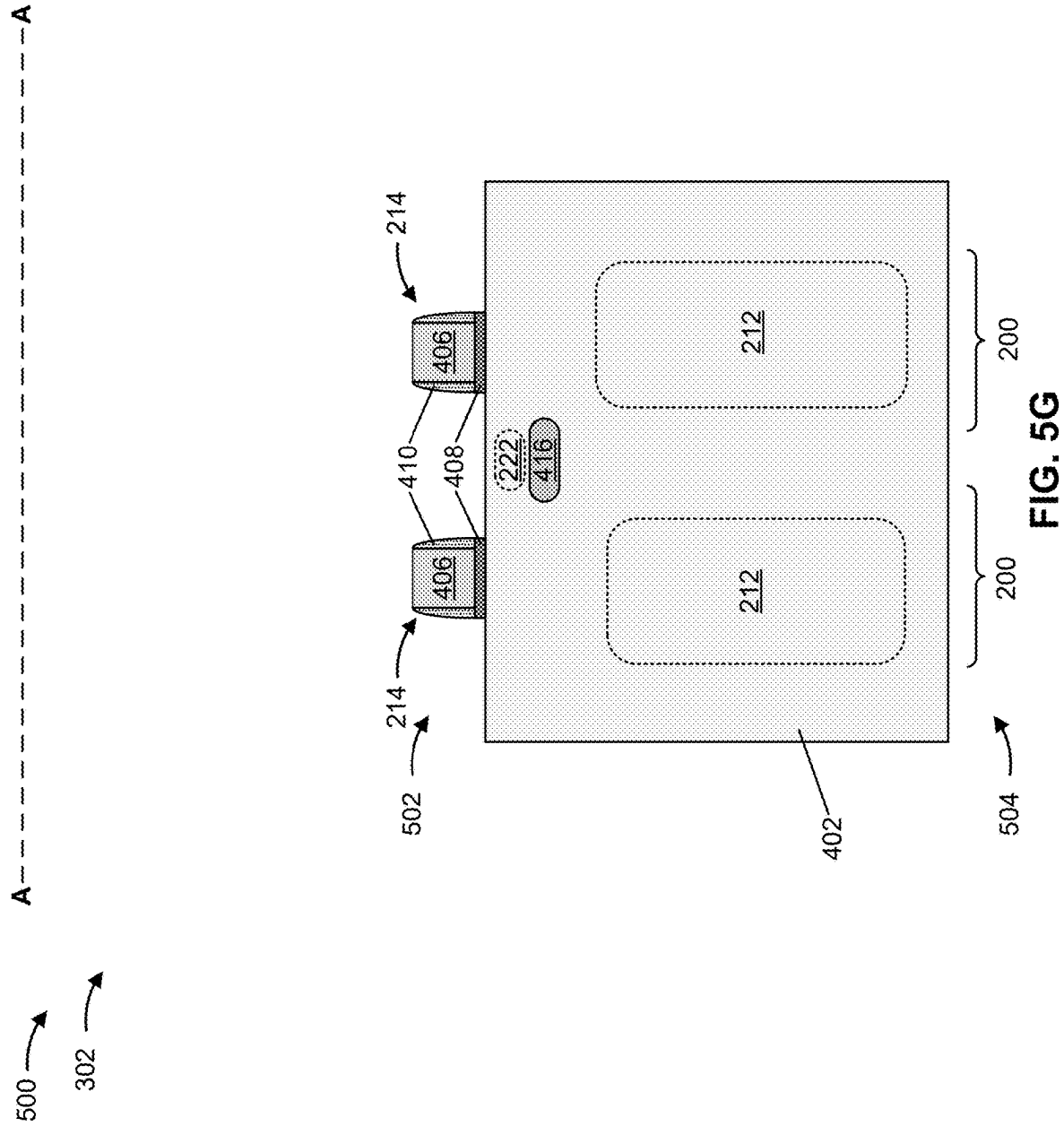
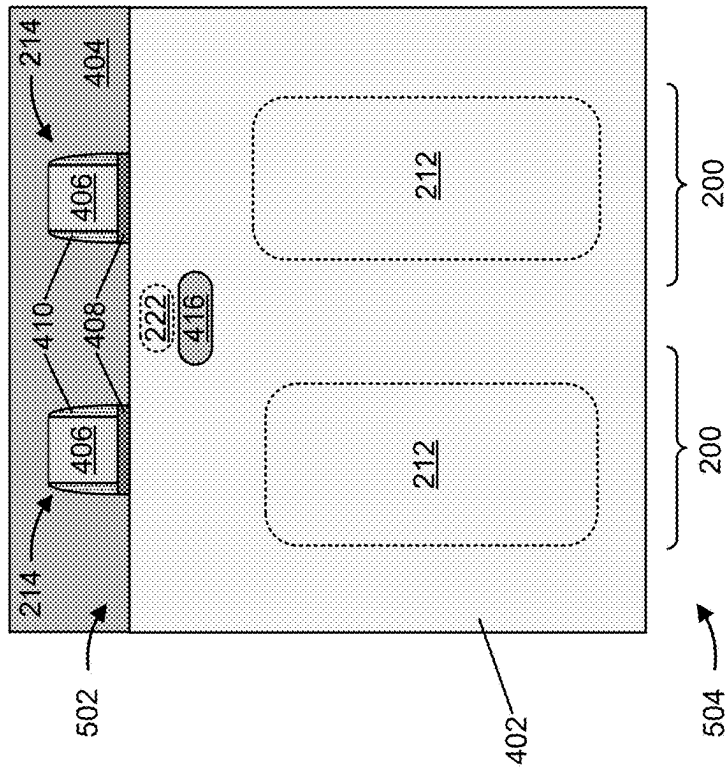


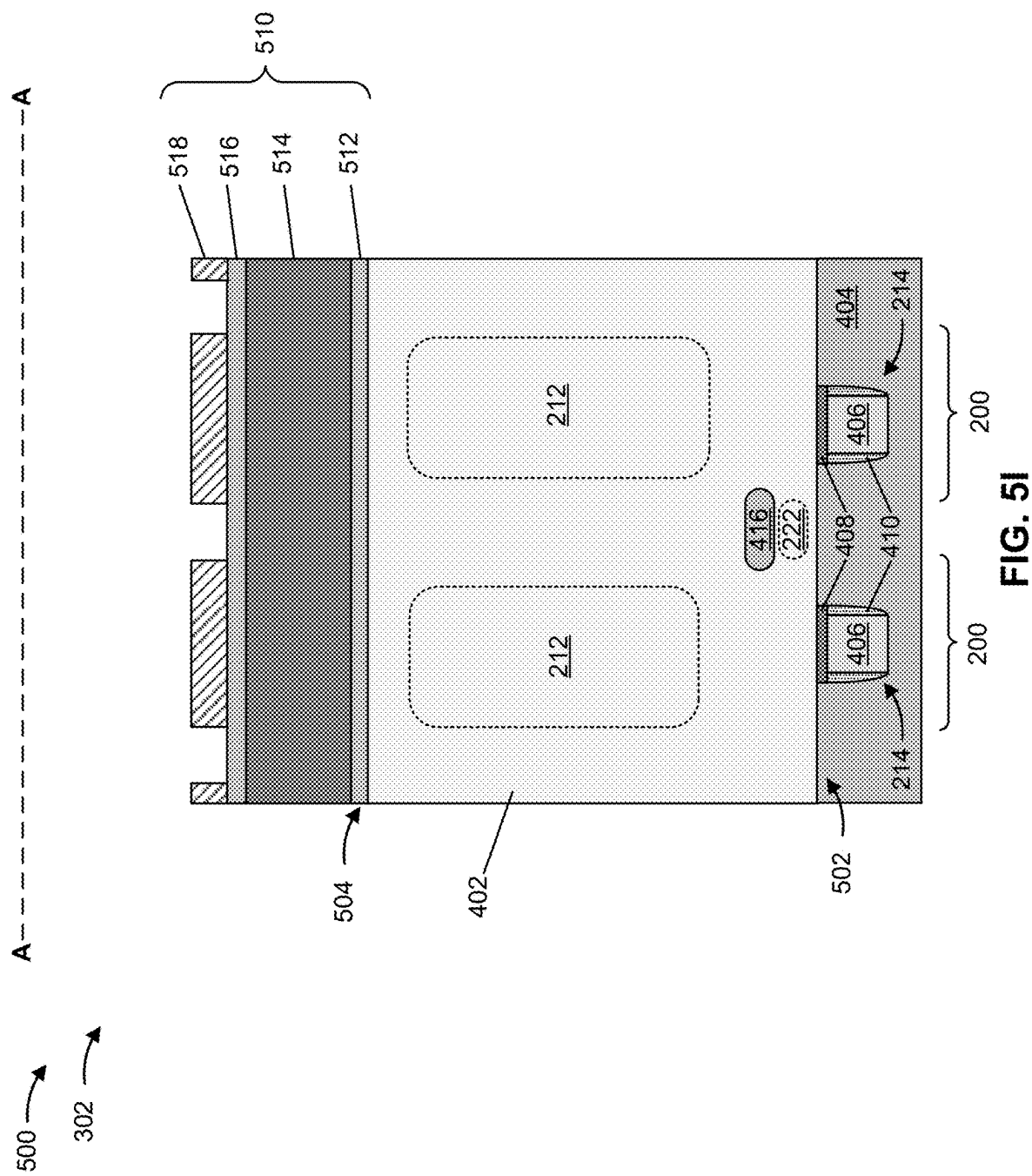
FIG. 5F



A-----A

500
302





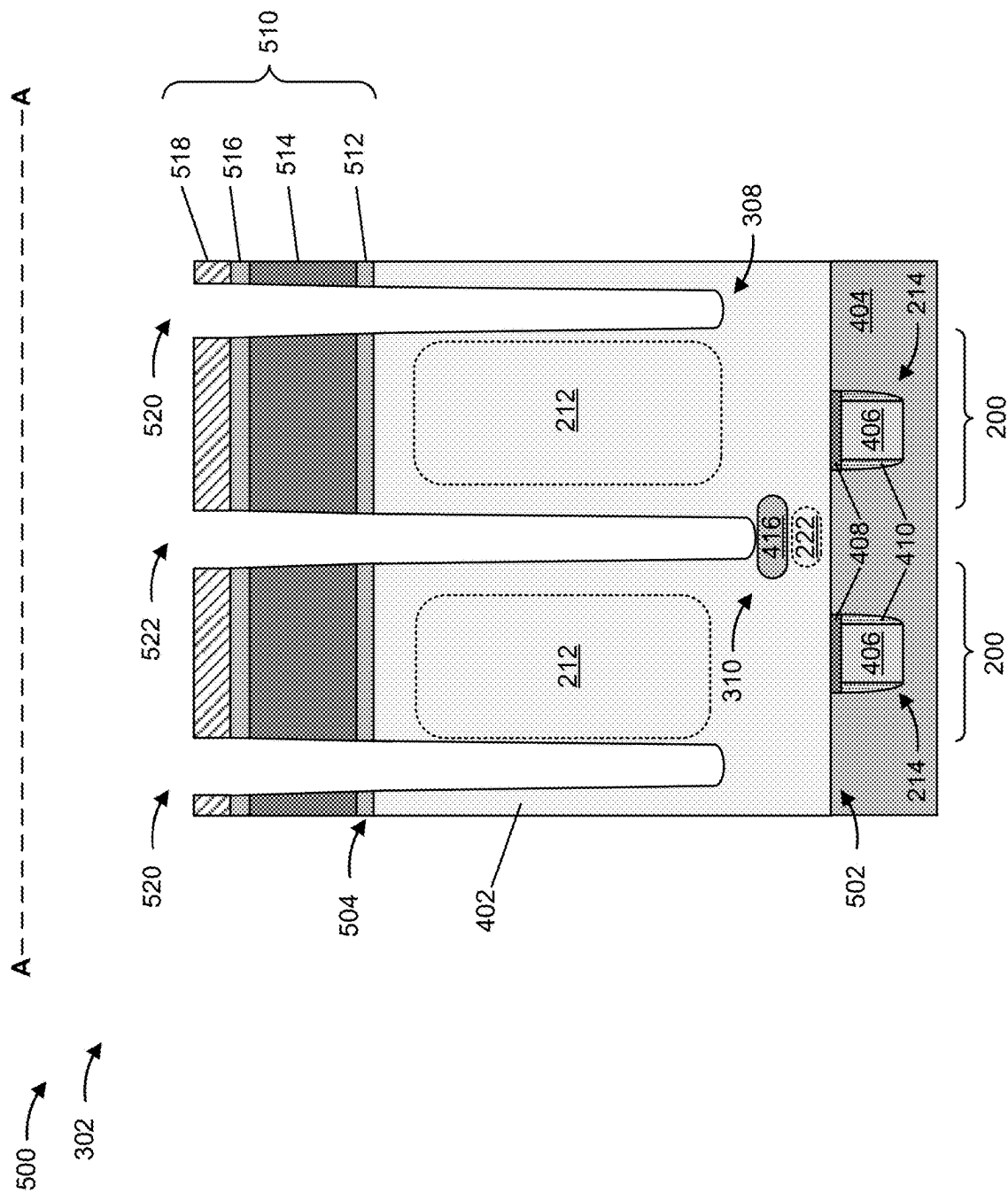
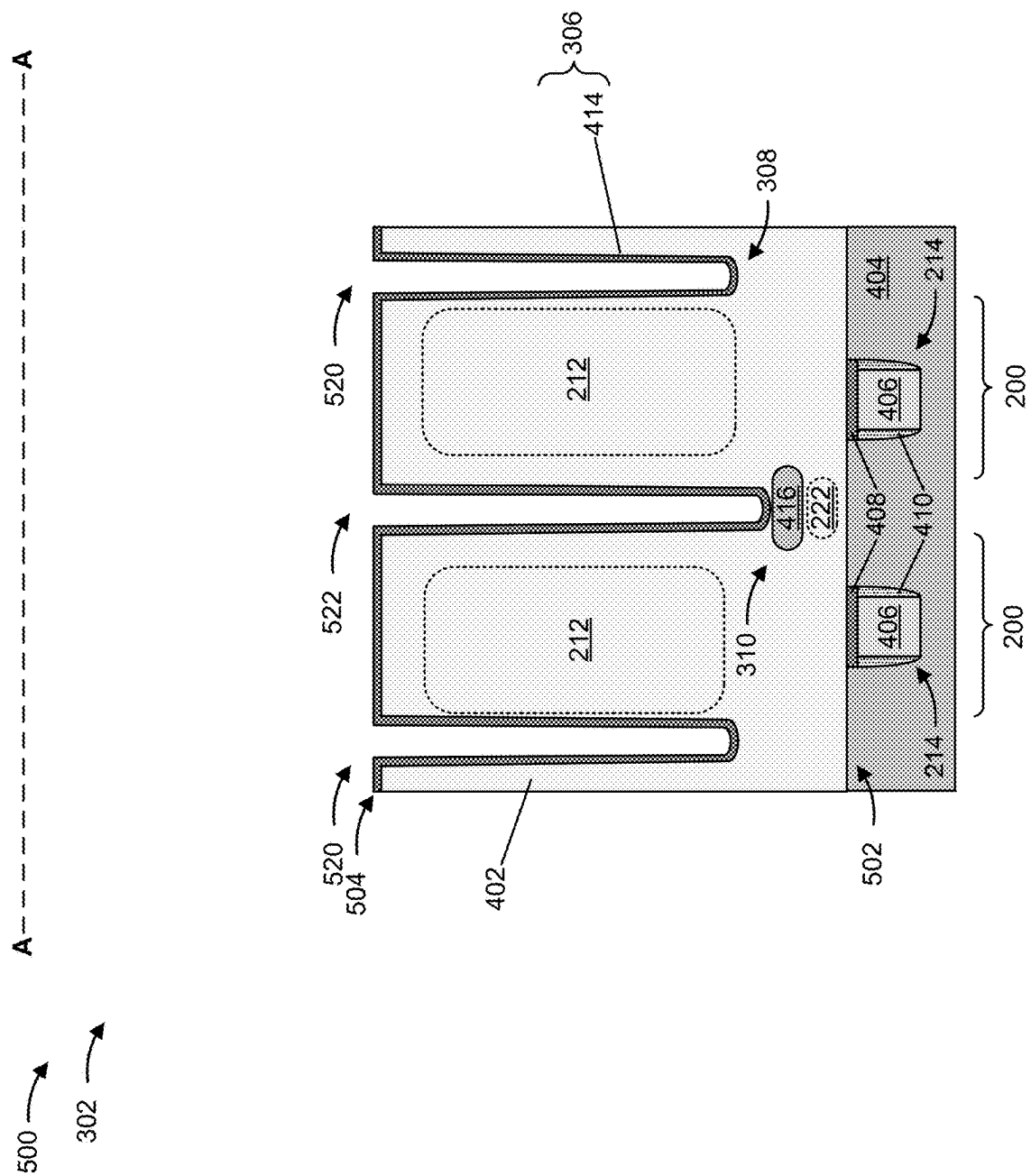
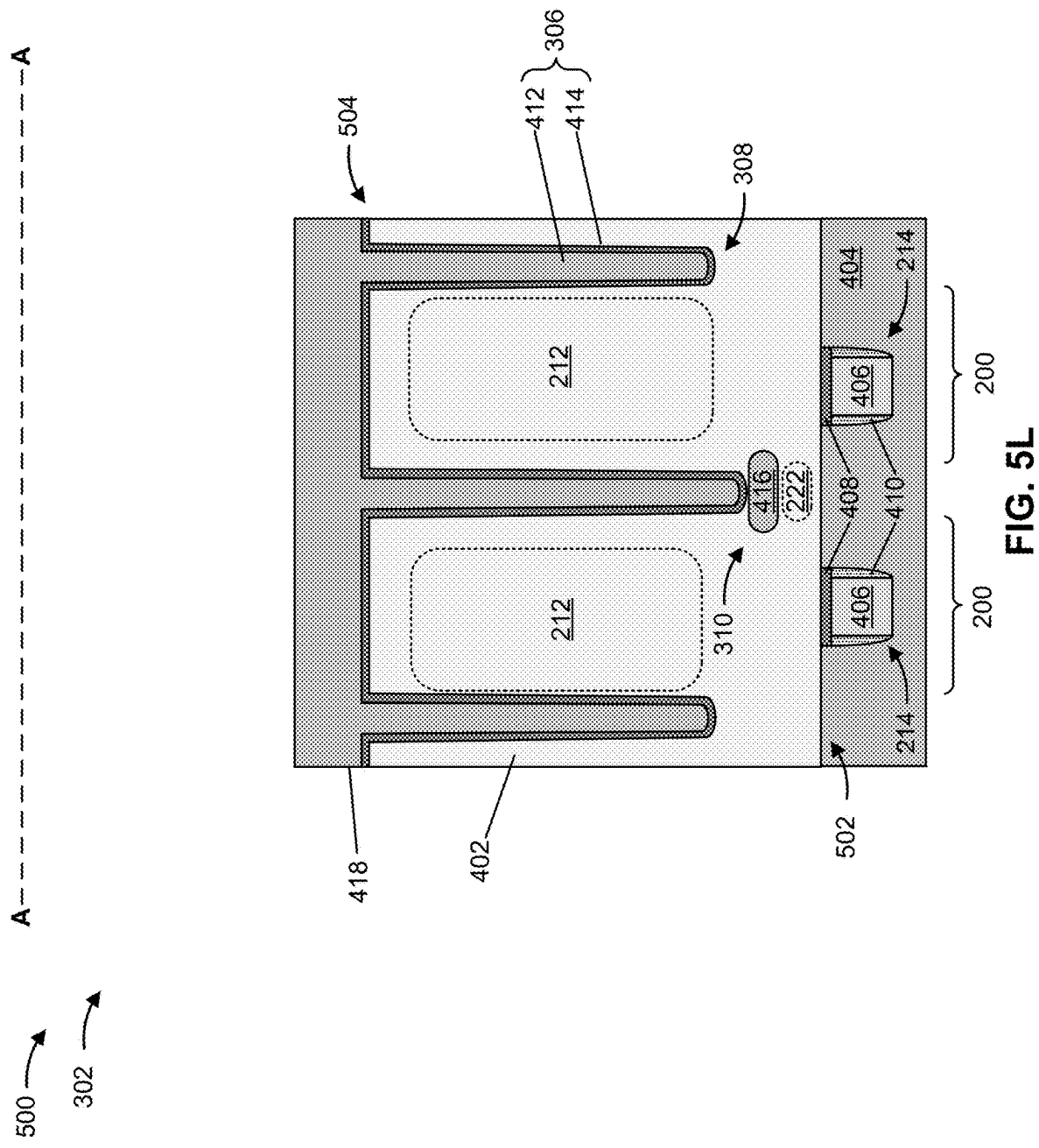


FIG. 5J





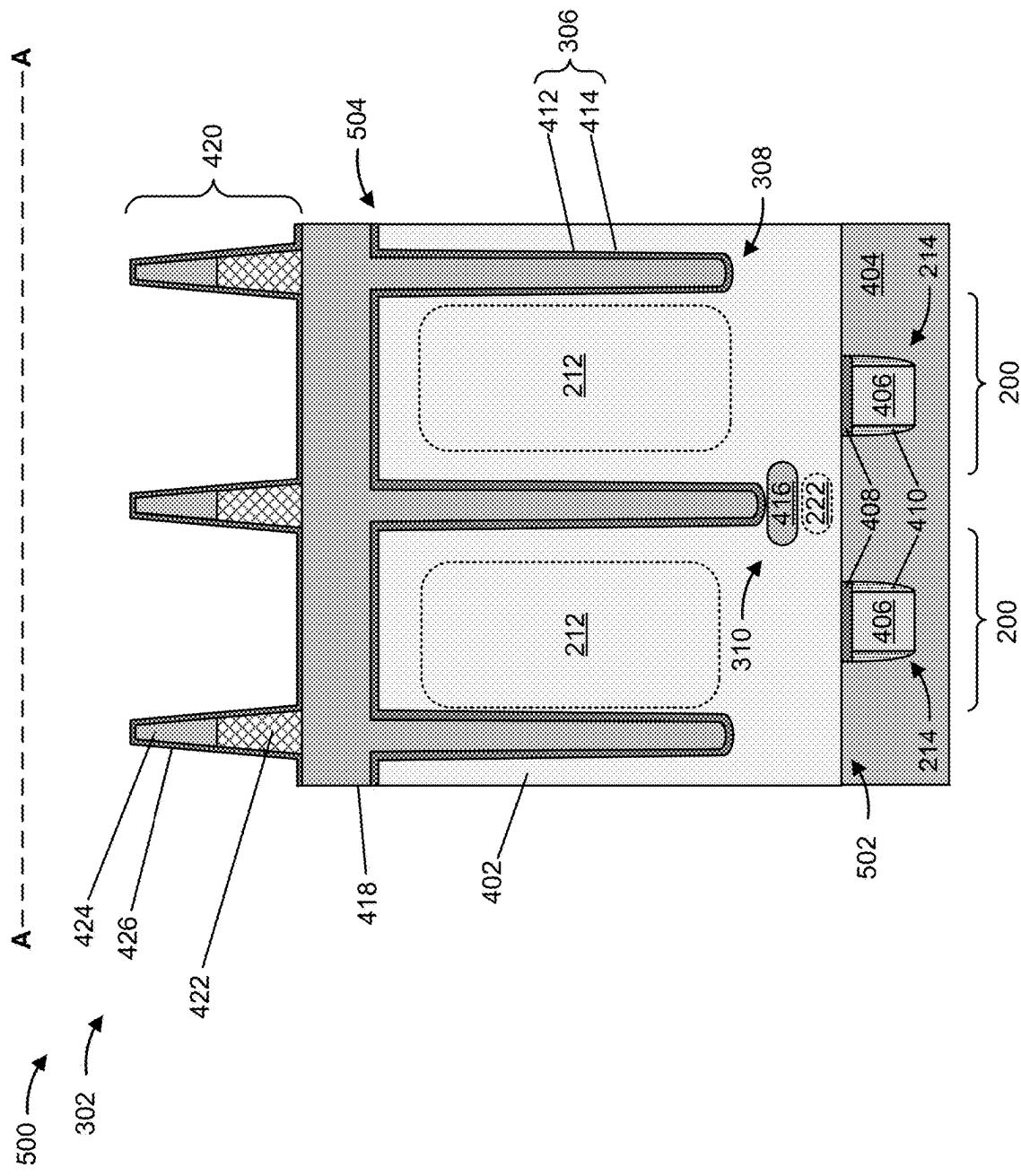


FIG. 5M

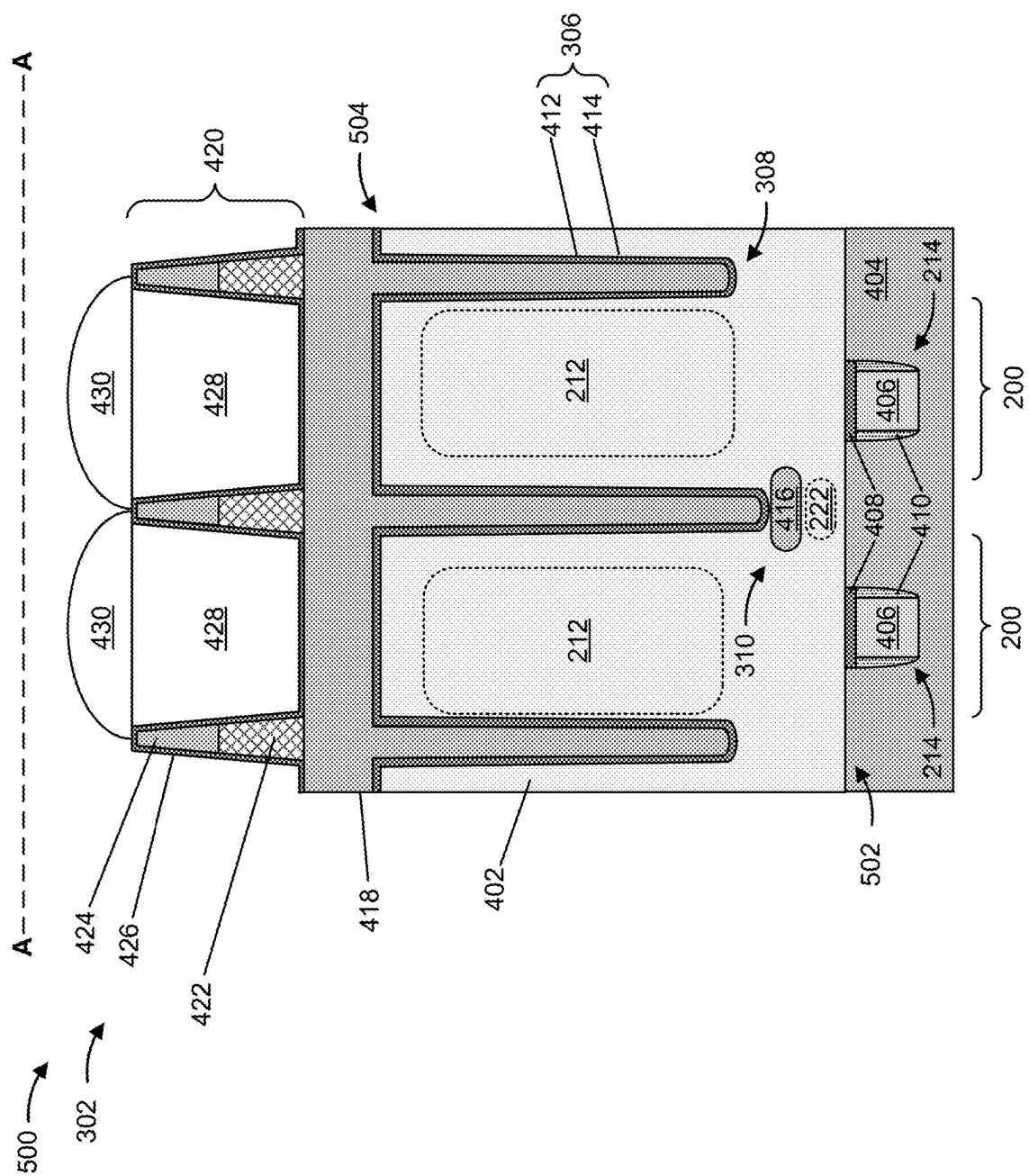
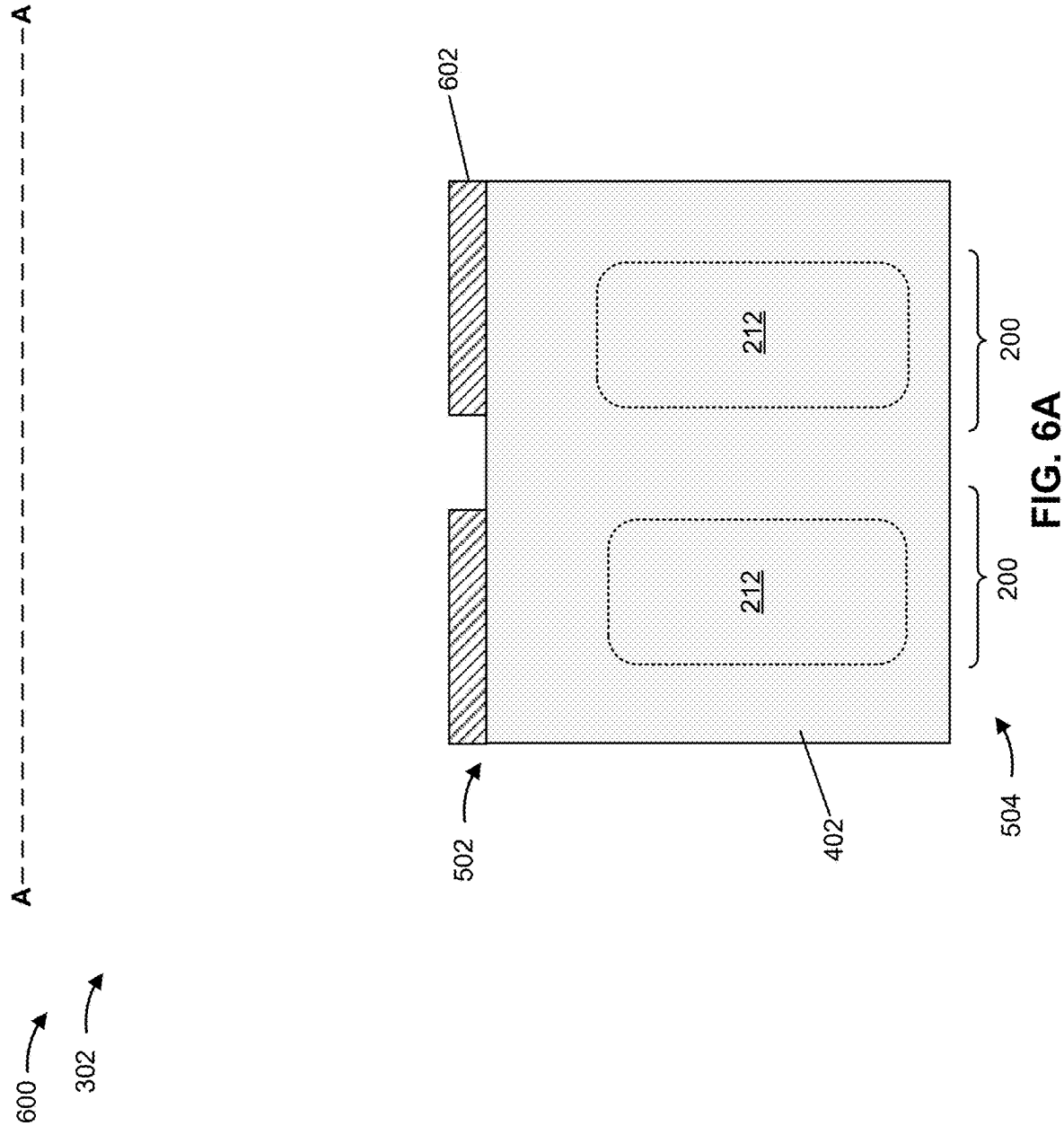
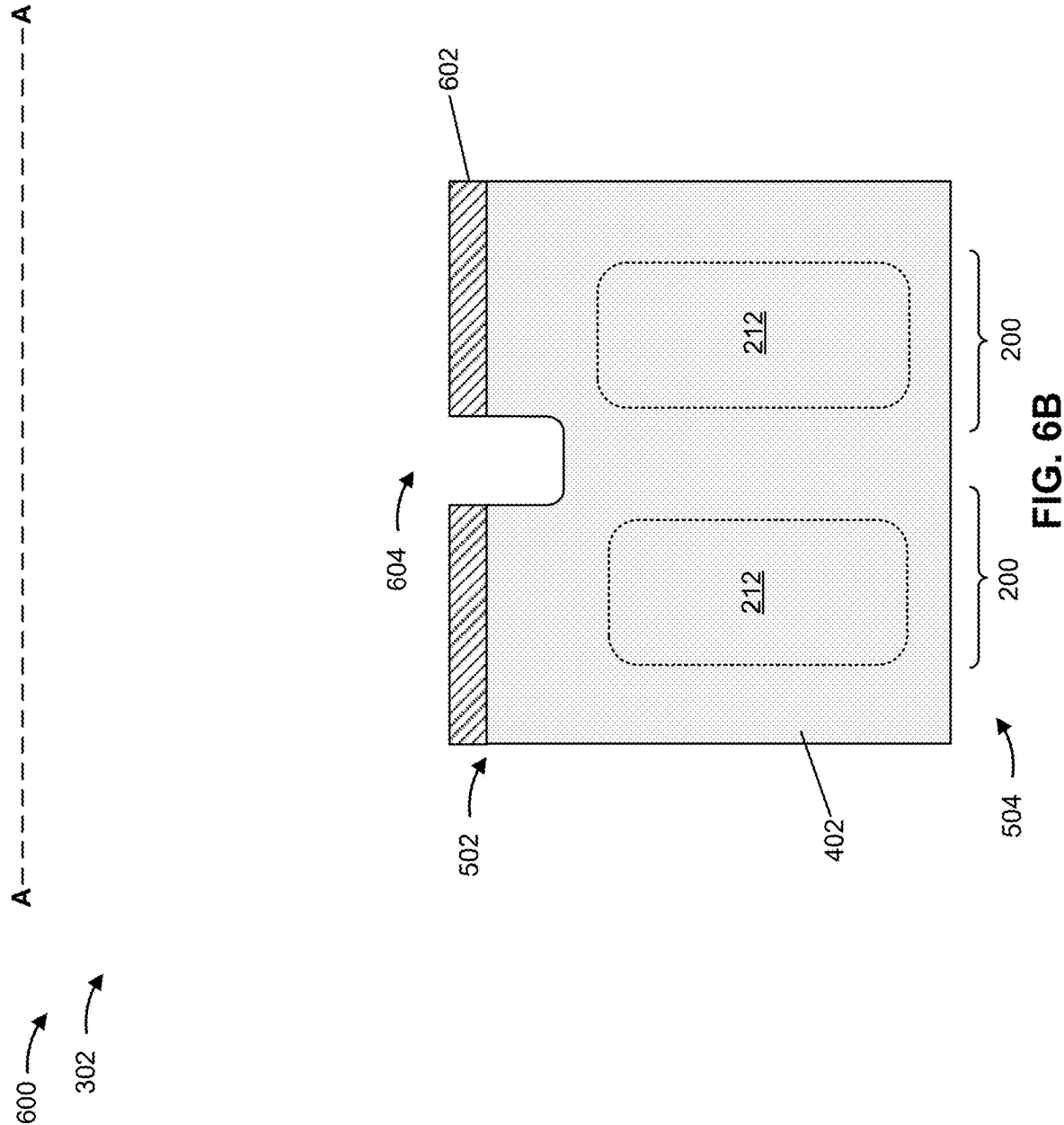
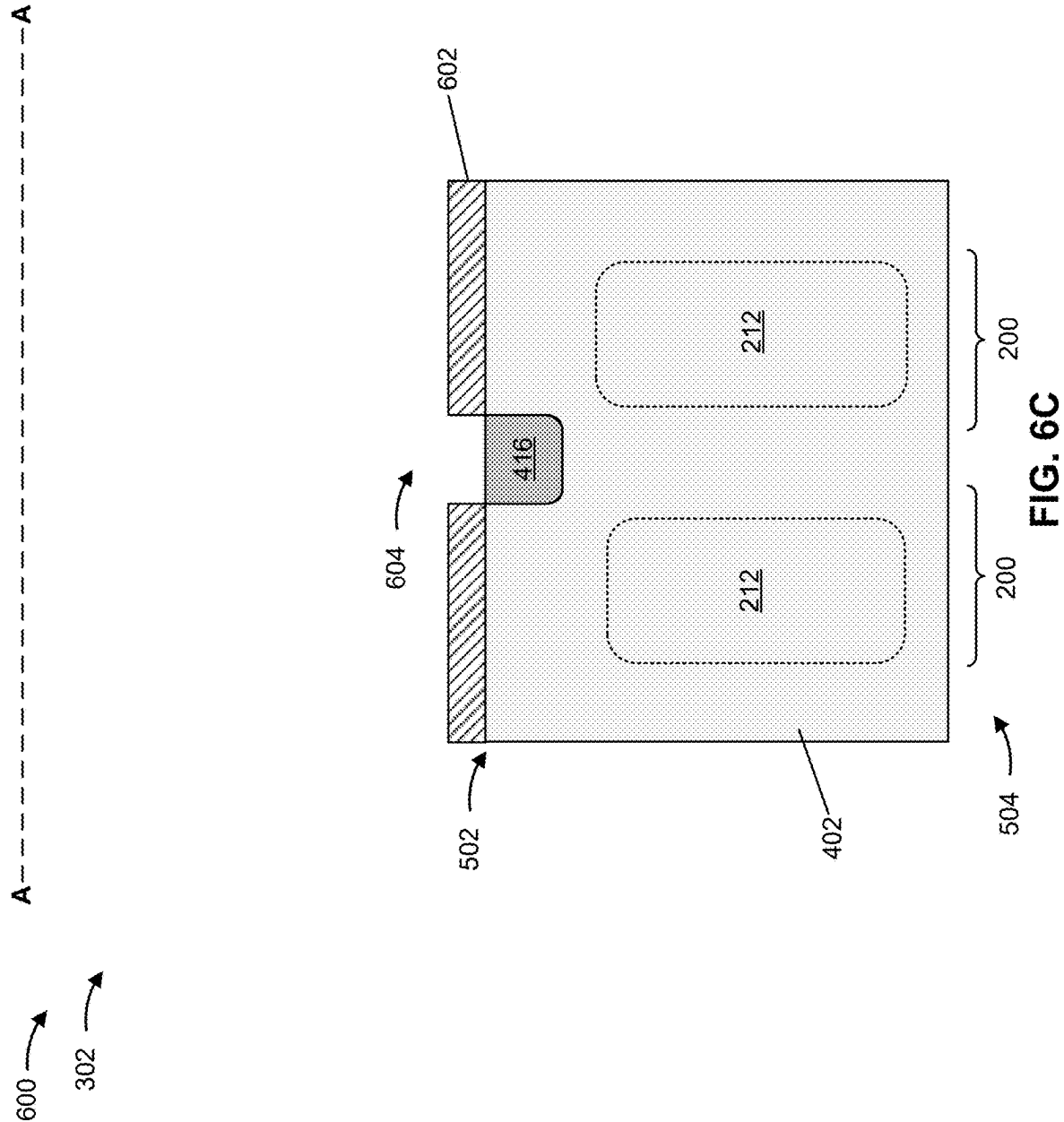
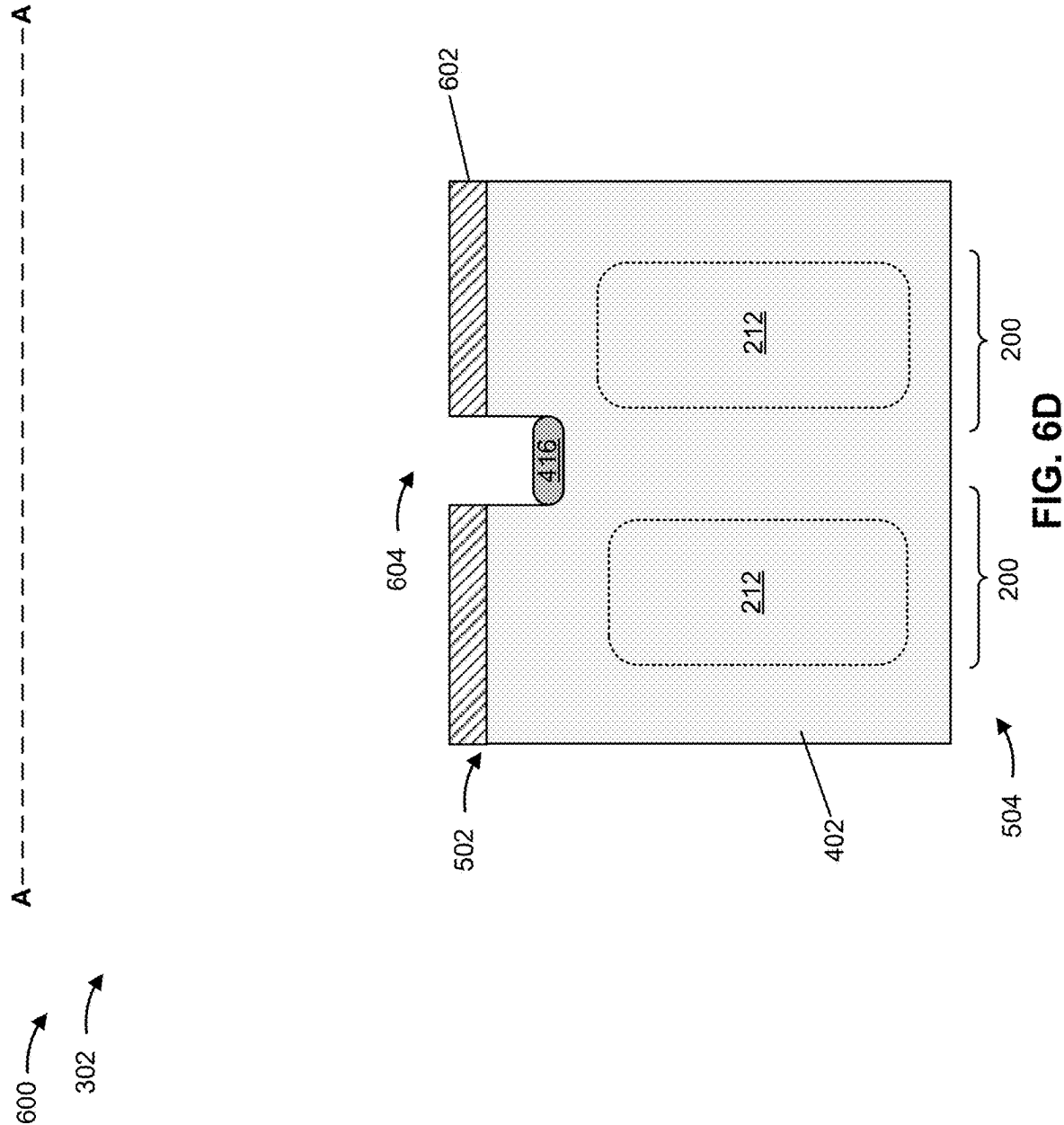


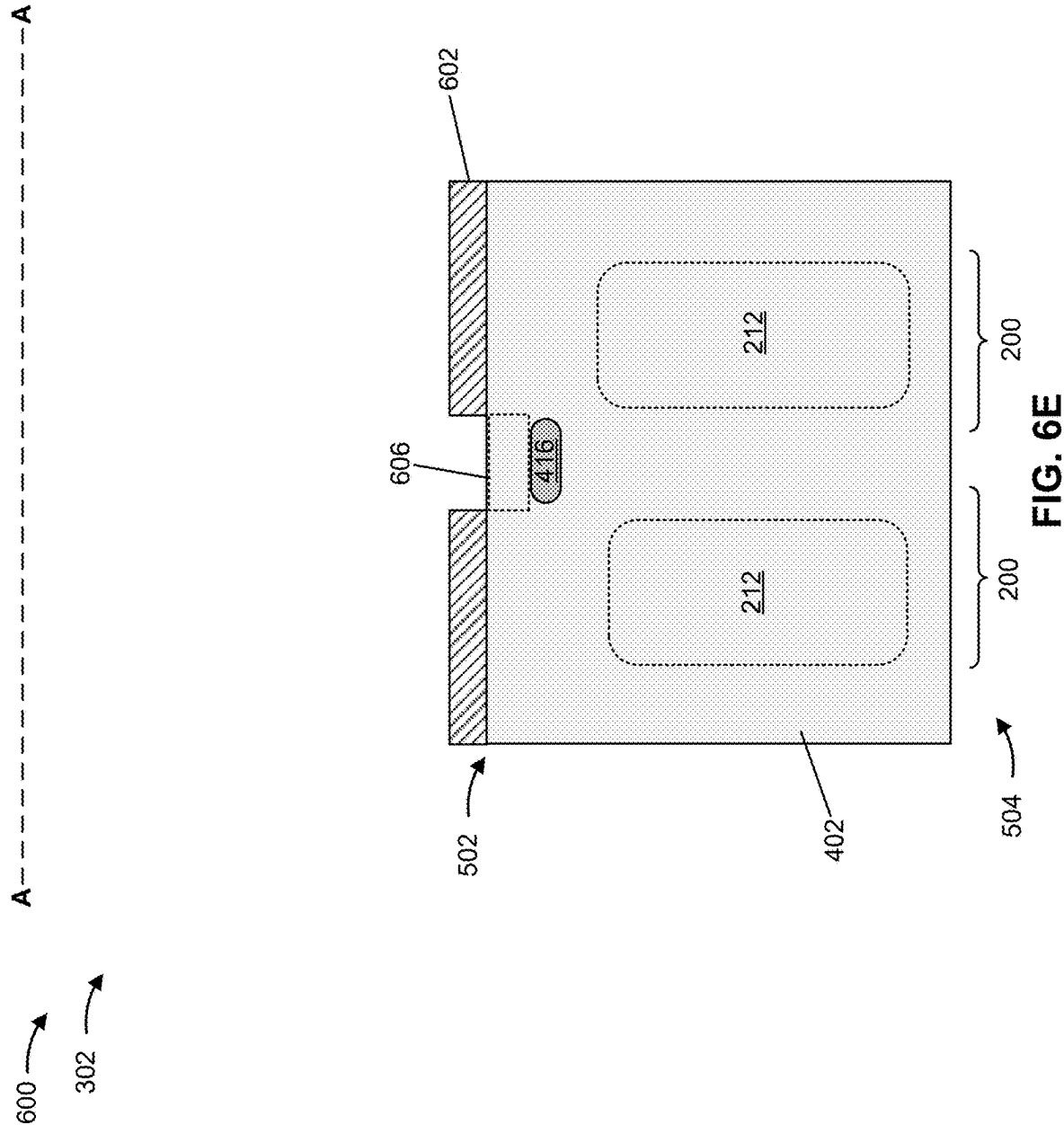
FIG. 5N

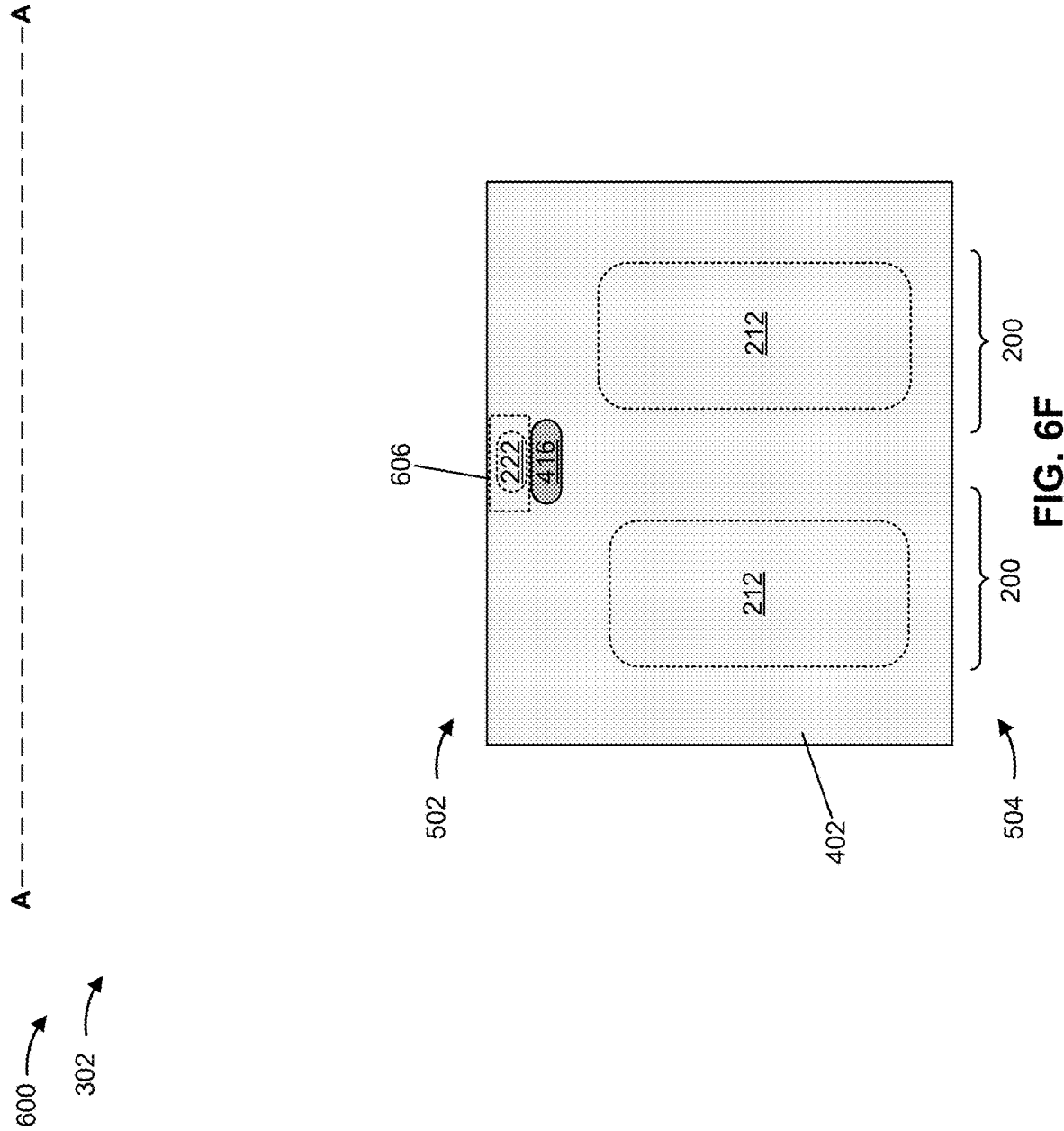












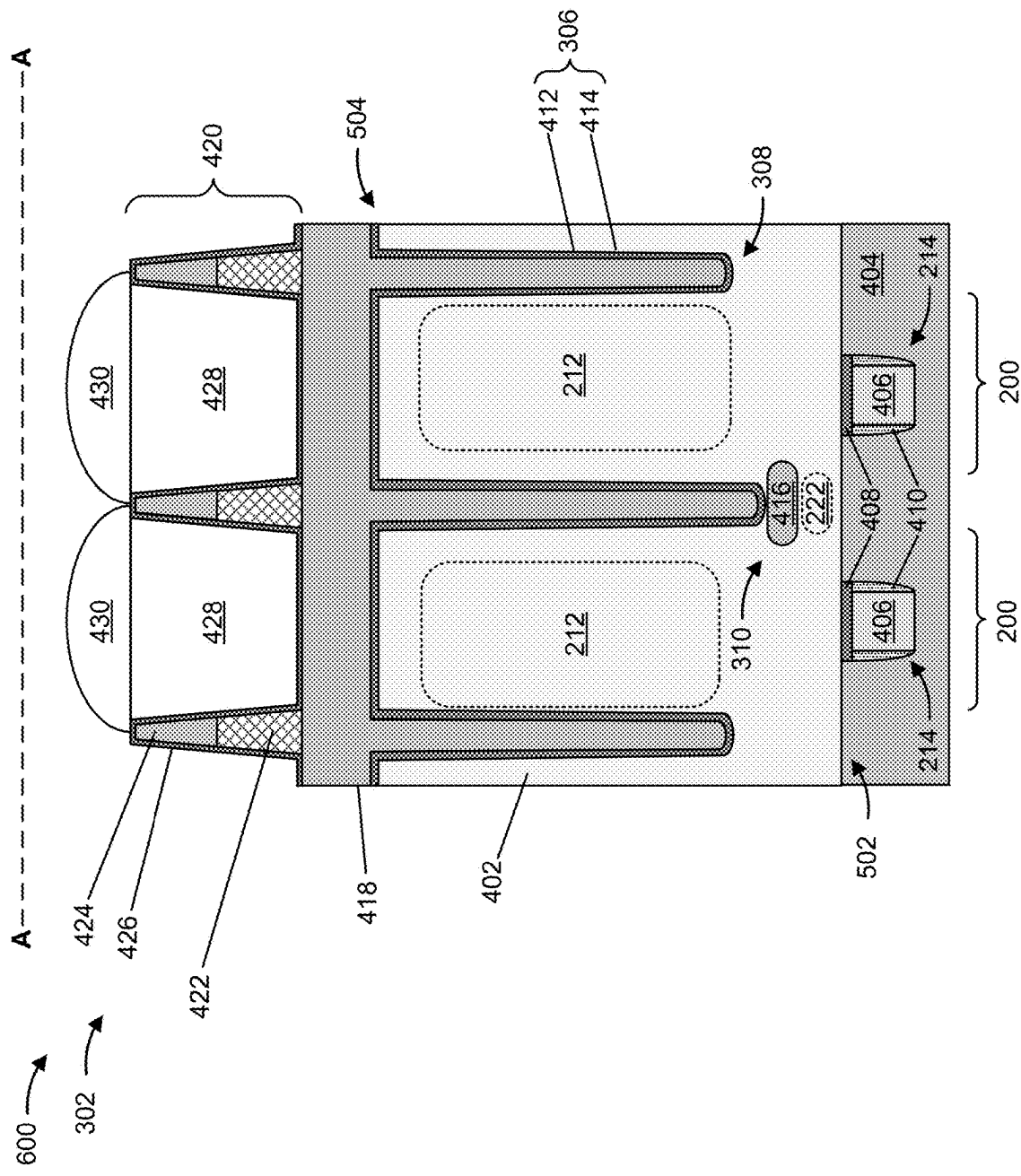


FIG. 6G

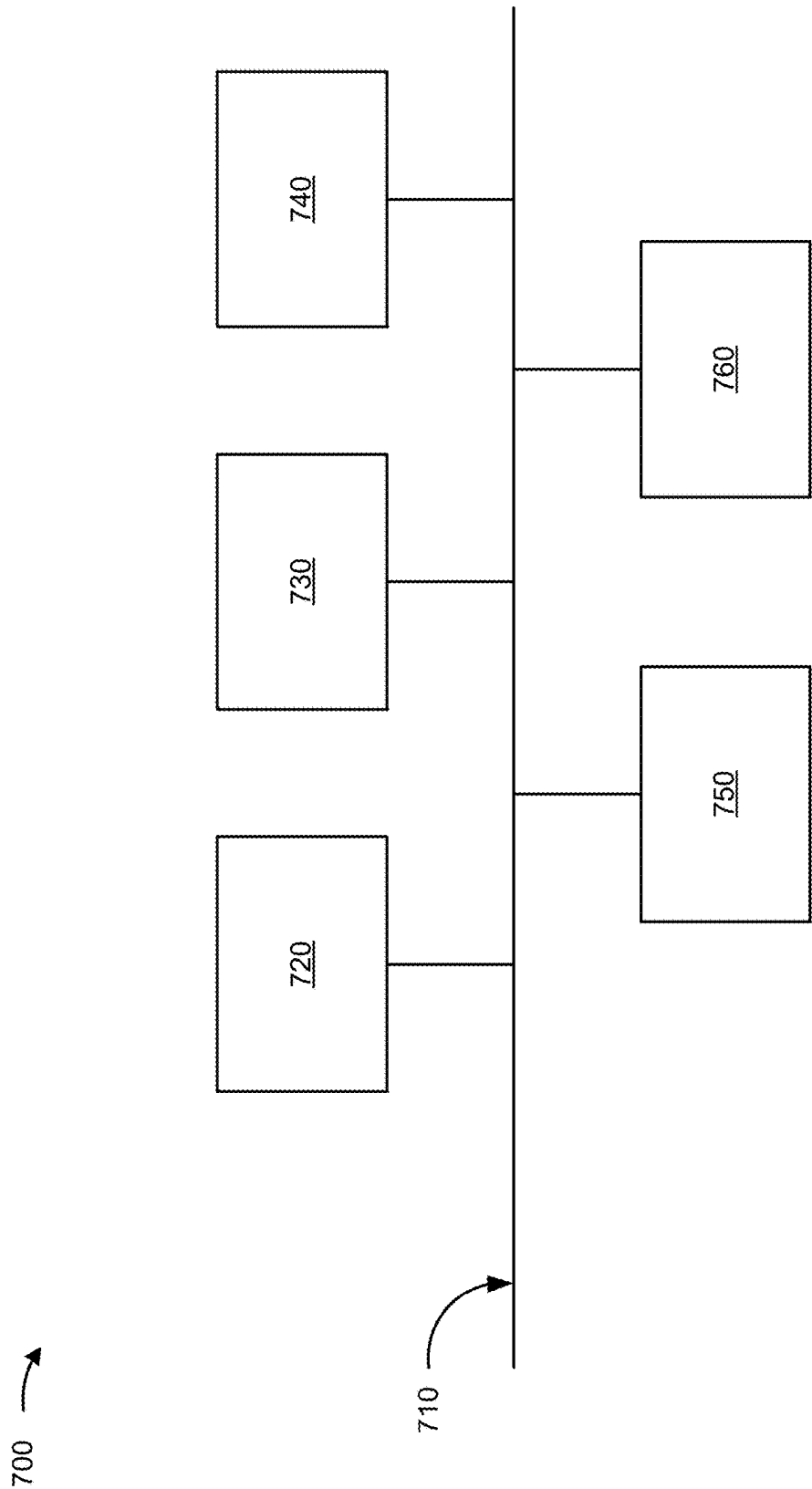


FIG. 7

800 →

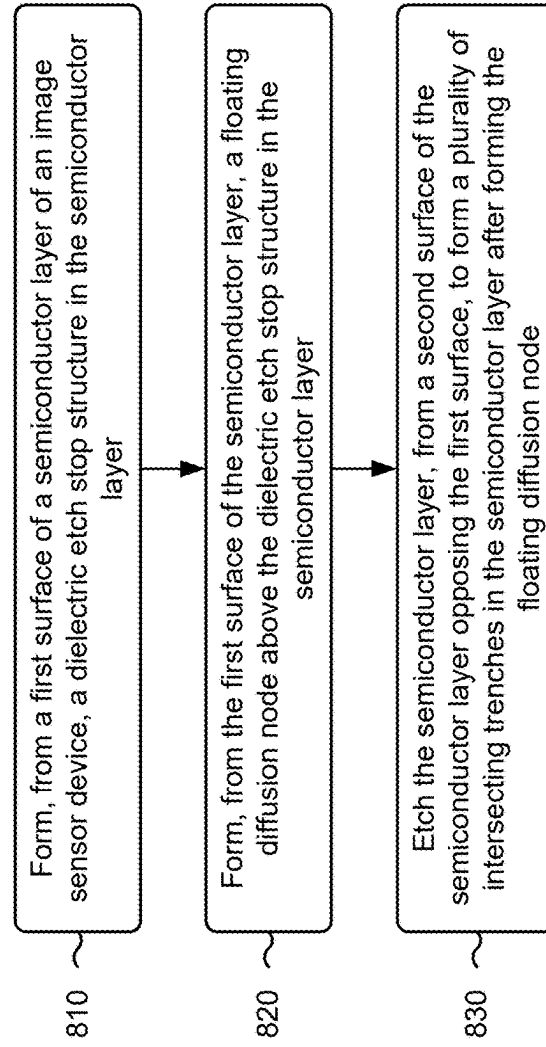


FIG. 8

900 →

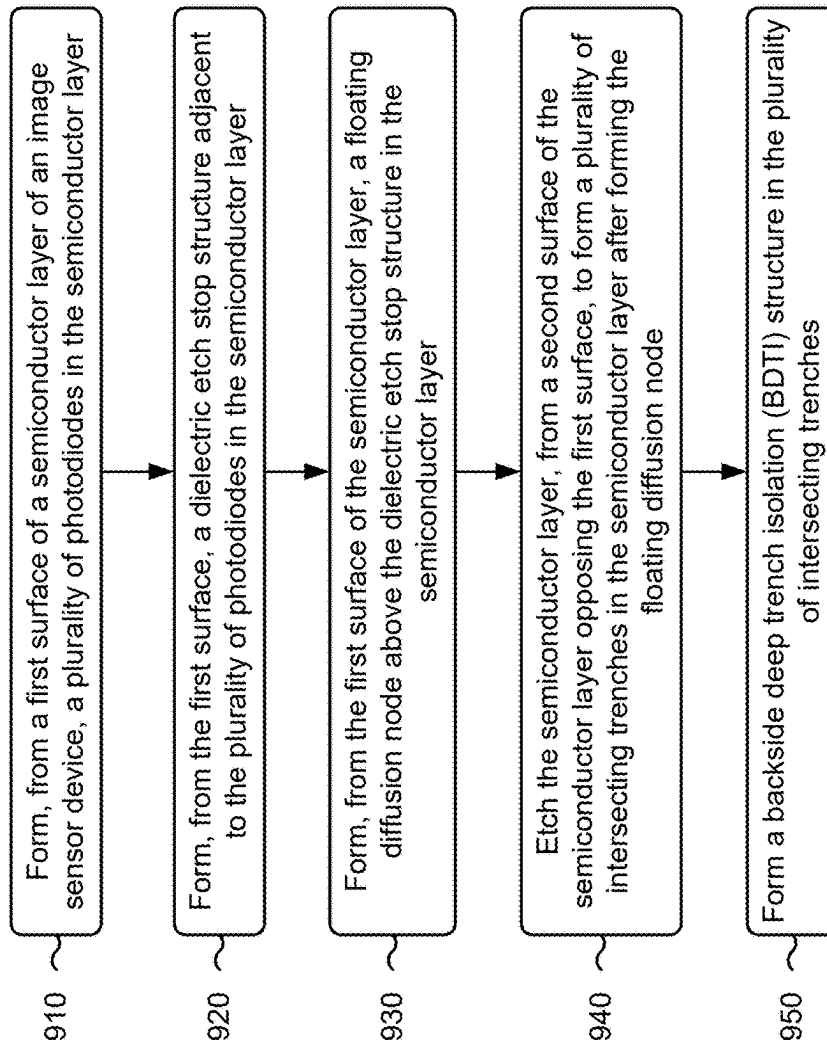


FIG. 9

IMAGE SENSOR DEVICE AND METHODS OF FORMATION

BACKGROUND

[0001] A complementary metal oxide semiconductor (CMOS) image sensor (CIS) may include a plurality of pixel sensors. A pixel sensor of the CMOS image sensor may include a transfer gate transistor, which may include a photodiode configured to convert photons of incident light into a photocurrent of electrons and a transfer gate configured to control the flow of the photocurrent between the photodiode and a drain region. The drain region may be configured to receive the photocurrent such that the photocurrent can be measured and/or transferred to other areas of the CMOS image sensor.

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. 1 is a diagram of an example environment in which systems and/or methods described herein may be implemented.

[0004] FIG. 2 is a diagram of an example of a pixel sensor described herein.

[0005] FIGS. 3A and 3B are diagrams of an example implementation of a pixel sensor array described herein.

[0006] FIGS. 4A and 4B are diagrams of an example implementation of a portion of a pixel sensor array described herein.

[0007] FIGS. 5A-5N are diagrams of an example implementation of forming a pixel sensor array of the image sensor device described herein.

[0008] FIGS. 6A-6G are diagrams of an example implementation of forming a pixel sensor array of the image sensor device described herein.

[0009] FIG. 7 is a diagram of example components of a device described herein.

[0010] FIG. 8 is a flowchart of an example process associated with forming an image sensor device described herein.

[0011] FIG. 9 is a flowchart of an example process associated with forming an image sensor device described herein.

DETAILED DESCRIPTION

[0012] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may

repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0013] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0014] Pixel sensors in a pixel sensor array are often separated by trench isolation structures. For example, a backside deep trench isolation (BDTI) structure may electrically and/or optically isolate pixel sensors in a pixel sensor array. The BDTI structure may include a grid of intersecting trenches arranged around the pixel sensors. Some regions of the BDTI structure (referred to as non-cross-road portions or non-X-road portions) may border (or may be included between) one or two pixel sensors in the pixel sensor array. Other regions (referred to as cross-road portions or X-road portions) of the BDTI structure may be located at an intersection between corners of three or more pixel sensors in the pixel sensor array.

[0015] An etch rate for etching recesses for the BDTI structure at cross-road portions generally may be greater relative to an etch rate for etching recesses at non-cross-road portions of the BDTI structures due to the greater spacing (sometimes referred to as “critical dimension” or “CD”) between the photodiodes in the cross-road portions than in the non-cross-road portions. The increase in spacing results from a greater area in which etchant is provided, resulting in a greater amount of etchant in the cross-road portions than in the non-cross-road portions. The greater amount of etchant results in the greater etch rate in the cross-road portions than in the non-cross-road portions, resulting in increased trench depth loading in the cross-road portions. “Trench depth loading” refers to a greater depth of etching in the cross-road portions than in the non-cross-road portions. The increased trench depth loading may increase the likelihood of etching into floating diffusion nodes in the cross-road portion, which may damage the floating diffusion nodes of the pixel sensors and result in failure of the pixel sensors.

[0016] In some implementations described herein, dielectric regions are formed in a semiconductor substrate of a semiconductor device. The dielectric regions may be formed in locations above which cross-road portions of a BDTI structure are to be formed for a pixel sensor array of the semiconductor device. The dielectric regions may function as etch stop structures in the cross-road portions when etching the semiconductor substrate to form the trenches in which the BDTI structure is to be formed. In particular, the dielectric regions minimize and/or prevent the increase in trench depth of the trenches in the cross-road portions relative to the trench depth of the trenches in the non-cross-road portions.

[0017] In this way, the dielectric regions reduce and/or minimize trench depth loading in the cross-road portions of the trenches. Thus, the dielectric regions prevent, minimize,

and/or reduce the likelihood of etching into floating diffusion nodes of pixel sensors of the pixel sensor array, which reduces the likelihood of degraded performance and/or failure of the pixel sensors.

[0018] FIG. 1 is a diagram of an example environment **100** in which systems and/or methods described herein may be implemented. As shown in FIG. 1, environment **100** may include a plurality of semiconductor processing tools **102-114** and a wafer/die transport tool **116**. The plurality of semiconductor processing tools **102-114** may include a deposition tool **102**, an exposure tool **104**, a developer tool **106**, an etch tool **108**, a planarization tool **110**, a plating tool **112**, an ion implantation tool **114**, and/or another type of semiconductor processing tool. The tools included in example environment **100** may be included in a semiconductor clean room, a semiconductor foundry, a semiconductor processing facility, and/or manufacturing facility, among other examples.

[0019] The deposition tool **102** is a semiconductor processing tool that includes a semiconductor processing chamber and one or more devices capable of depositing various types of materials onto a substrate. In some implementations, the deposition tool **102** includes a spin coating tool that is capable of depositing a photoresist layer on a substrate such as a wafer. In some implementations, the deposition tool **102** includes a chemical vapor deposition (CVD) tool such as a plasma enhanced CVD (PECVD) tool, a low pressure CVD (LPCVD) tool, a high-density plasma CVD (HDP-CVD) tool, a sub-atmospheric CVD (SACVD) tool, an atomic layer deposition (ALD) tool, a plasma-enhanced atomic layer deposition (PEALD) tool, or another type of CVD tool. In some implementations, the deposition tool **102** includes a physical vapor deposition (PVD) tool, such as a sputtering tool or another type of PVD tool. In some implementations, the example environment **100** includes a plurality of types of deposition tools **102**.

[0020] The exposure tool **104** is a semiconductor processing tool that is capable of exposing a photoresist layer to a radiation source, such as an ultraviolet light (UV) source (e.g., a deep UV light source, an extreme UV light (EUV) source, and/or the like), an x-ray source, an electron beam (e-beam) source, and/or the like. The exposure tool **104** may expose a photoresist layer to the radiation source to transfer a pattern from a photomask to the photoresist layer. The pattern may include one or more semiconductor device layer patterns for forming one or more semiconductor devices, may include a pattern for forming one or more structures of a semiconductor device, may include a pattern for etching various portions of a semiconductor device, and/or the like. In some implementations, the exposure tool **104** includes a scanner, a stepper, or a similar type of exposure tool.

[0021] The developer tool **106** is a semiconductor processing tool that is capable of developing a photoresist layer that has been exposed to a radiation source to develop a pattern transferred to the photoresist layer from the exposure tool **104**. In some implementations, the developer tool **106** develops a pattern by removing unexposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by removing exposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by dissolving exposed or unexposed portions of a photoresist layer through the use of a chemical developer.

[0022] The etch tool **108** is a semiconductor processing tool that is capable of etching various types of materials of a substrate, wafer, or semiconductor device. For example, the etch tool **108** may include a wet etch tool, a dry etch tool, and/or the like. In some implementations, the etch tool **108** includes a chamber that is filled with an etchant, and the substrate is placed in the chamber for a particular time period to remove particular amounts of one or more portions of the substrate. In some implementations, the etch tool **108** may etch one or more portions of the substrate using a plasma etch or a plasma-assisted etch, which may involve using an ionized gas to isotropically or directionally etch the one or more portions.

[0023] The planarization tool **110** is a semiconductor processing tool that is capable of polishing or planarizing various layers of a wafer or semiconductor device. For example, a planarization tool **110** may include a chemical mechanical planarization (CMP) tool and/or another type of planarization tool that polishes or planarizes a layer or surface of deposited or plated material. The planarization tool **110** may polish or planarize a surface of a semiconductor device with a combination of chemical and mechanical forces (e.g., chemical etching and free abrasive polishing). The planarization tool **110** may utilize an abrasive and corrosive chemical slurry in conjunction with a polishing pad and retaining ring (e.g., typically of a greater diameter than the semiconductor device). The polishing pad and the semiconductor device may be pressed together by a dynamic polishing head and held in place by the retaining ring. The dynamic polishing head may rotate with different axes of rotation to remove material and even out any irregular topography of the semiconductor device, making the semiconductor device flat or planar.

[0024] The plating tool **112** is a semiconductor processing tool that is capable of plating a substrate (e.g., a wafer, a semiconductor device, and/or the like) or a portion thereof with one or more metals. For example, the plating tool **112** may include a copper electroplating device, an aluminum electroplating device, a nickel electroplating device, a tin electroplating device, a compound material or alloy (e.g., tin-silver, tin-lead, and/or the like) electroplating device, and/or an electroplating device for one or more other types of conductive materials, metals, and/or similar types of materials.

[0025] The ion implantation tool **114** is a semiconductor processing tool that is capable of implanting ions into a substrate. The ion implantation tool **114** may generate ions in an arc chamber from a source material such as a gas or a solid. The source material may be provided into the arc chamber, and an arc voltage is discharged between a cathode and an electrode to produce a plasma containing ions of the source material. One or more extraction electrodes may be used to extract the ions from the plasma in the arc chamber and accelerate the ions to form an ion beam. The ion beam may be directed toward the substrate such that the ions are implanted below the surface of the substrate.

[0026] The wafer/die transport tool **116** may be included in a cluster tool or another type of tool that includes a plurality of processing chambers, and may be configured to transport substrates and/or semiconductor devices between the plurality of processing chambers, to transport substrates and/or semiconductor devices between a processing chamber and a buffer area, to transport substrates and/or semiconductor devices between a processing chamber and an

interface tool such as an equipment front end module (EFEM), and/or to transport substrates and/or semiconductor devices between a processing chamber and a transport carrier (e.g., a front opening unified pod (FOUP)), among other examples. In some implementations, a wafer/die transport tool **116** may be included in a multi-chamber (or cluster) deposition tool **102**, which may include a pre-clean processing chamber (e.g., for cleaning or removing oxides, oxidation, and/or other types of contamination or byproducts from a substrate and/or semiconductor device) and a plurality of types of deposition processing chambers (e.g., processing chambers for depositing different types of materials, processing chambers for performing different types of deposition operations).

[0027] In some implementations, one or more of the semiconductor processing tools **102-114** and/or the wafer/die transport tool **116** may be used to perform one or more semiconductor processing operations described herein. For example, one or more of the semiconductor processing tools **102-114** and/or the wafer/die transport tool **116** may be used to form, from a first surface of a semiconductor layer of an image sensor device, a dielectric etch stop structure in the semiconductor layer; form, from the first surface of the semiconductor layer, a floating diffusion node above the dielectric etch stop structure in the semiconductor layer; and/or etch the semiconductor layer, from a second surface of the semiconductor layer opposing the first surface, to form a plurality of intersecting trenches in the semiconductor layer after forming the floating diffusion node, where two or more of the intersecting trenches intersect in a cross-road portion above the dielectric etch stop structure and the floating diffusion node, and where the etching of the semiconductor layer stops on the dielectric etch stop structure in the cross-road portion, among other examples.

[0028] As another example, one or more of the semiconductor processing tools **102-114** and/or the wafer/die transport tool **116** may be used to form, from a first surface of a semiconductor layer of an image sensor device, a plurality of photodiodes in the semiconductor layer; form, from the first surface, a dielectric etch stop structure adjacent to the plurality of photodiodes in the semiconductor layer, where a distance between the dielectric etch stop structure and the first surface is less than a distance between the plurality of photodiodes and the first surface; form, from the first surface of the semiconductor layer, a floating diffusion node above the dielectric etch stop structure in the semiconductor layer; etch the semiconductor layer, from a second surface of the semiconductor layer opposing the first surface, to form a plurality of intersecting trenches in the semiconductor layer after forming the floating diffusion node, where two or more of the intersecting trenches intersect in a cross-road portion above the dielectric etch stop structure and the floating diffusion node, and where the etching of the semiconductor layer stops on the dielectric etch stop structure in the cross-road portion; and/or form a BDTI structure in the plurality of intersecting trenches, where a portion of the BDTI structure, in the cross-road portion, is formed on the dielectric etch stop structure, among other examples.

[0029] In some implementations, one or more of the semiconductor processing tools **102-114** and/or the wafer/die transport tool **116** may be used to perform other semiconductor processing operations described herein, such as in connection with FIGS. **5A-5N**, **6A-6G**, **8**, and/or **9**, among other examples.

[0030] The number and arrangement of devices shown in FIG. **1** are provided as one or more examples. In practice, there may be additional devices, fewer devices, different devices, or differently arranged devices than those shown in FIG. **1**. Furthermore, two or more devices shown in FIG. **1** may be implemented within a single device, or a single device shown in FIG. **1** may be implemented as multiple, distributed devices. Additionally, or alternatively, a set of devices (e.g., one or more devices) of the example environment **100** may perform one or more functions described as being performed by another set of devices of the example environment **100**.

[0031] FIG. **2** is a diagram of an example of a pixel sensor **200** described herein. The pixel sensor **200** may include a front side pixel sensor (e.g., a pixel sensor that is configured to receive photons of light from a front side of a sensor die), a back side pixel sensor (e.g., a pixel sensor that is configured to receive photons of light from a back side of a sensor die), and/or another type of pixel sensor. The pixel sensor **200** may be electrically connected to a supply voltage (V_{dd}) **202** and an electrical ground **204**.

[0032] The pixel sensor **200** includes a sensing region **206** that may be configured to sense and/or accumulate incident light (e.g., light directed toward the pixel sensor **200**). The pixel sensor **200** also includes a control circuitry region **208**. The control circuitry region **208** is electrically connected with the sensing region **206** and is configured to receive a photocurrent **210** that is generated by the sensing region **206**. Moreover, the control circuitry region **208** is configured to transfer the photocurrent **210** from the sensing region **206** to downstream circuits such as amplifiers or analog-to-digital (AD) converters, among other examples.

[0033] The sensing region **206** includes a photodiode **212**. The photodiode **212** may absorb and accumulate photons of the incident light, and may generate the photocurrent **210** based on absorbed photons. The magnitude of the photocurrent **210** is based on the amount of light collected in the photodiode **212**. Thus, the accumulation of photons in the photodiode **212** generates a build-up of electrical charge that represents the intensity or brightness of the incident light (e.g., a greater amount of charge may correspond to a greater intensity or brightness, and a lower amount of charge may correspond to a lower intensity or brightness).

[0034] The photodiode **212** is electrically connected with a source of a transfer gate **214** in the control circuitry region **208**. The transfer gate **214** is configured to control the transfer of the photocurrent **210** from the photodiode **212**. The photocurrent **210** is provided from the source of the transfer gate **214** to a drain of the transfer gate **214** based on selectively switching a gate of the transfer gate **214**. The gate of the transfer gate **214** may be selectively switched by applying a transfer voltage (V_{tr}) **216** to the transfer gate **214**. In some implementations, the transfer voltage **216** being applied to the transfer gate **214** causes a conductive channel to form between the source and the drain of the transfer gate **214**, which enables the photocurrent **210** to traverse along the conductive channel from the source to the drain. In some implementations, the transfer voltage **216** being removed from the transfer gate **214** (or the absence of the transfer voltage **216**) causes the conductive channel to be removed such that the photocurrent **210** cannot pass from the source to the drain.

[0035] The control circuitry region **208** further includes a reset gate **218**. The reset gate **218** is electrically connected

to the supply voltage **202**. The reset gate **218** may be controlled by a reset voltage (V_{rst}) **220**. The transfer gate **214** and the reset gate **218** may be electrically coupled with a floating diffusion node **222**. The reset voltage **220** may be applied to the reset gate **218** to pull the drain of the transfer gate **214** to a high voltage (e.g., to the supply voltage **202**) to “reset” the floating diffusion node **222** (e.g., by draining any residual charge in the floating diffusion node **222**) prior to activation of the transfer gate **214** to transfer the photocurrent **210** from the photodiode **212** to the floating diffusion node **222**.

[0036] The photocurrent **210** may be used to apply a floating diffusion voltage (V_{fd}) to a source follower gate **224** of the control circuitry region **208**. This permits the photocurrent **210** to be observed without removing or discharging the photocurrent **210** from the floating diffusion node **222**. The reset gate **218** may instead be used to remove or discharge the photocurrent **210** from the floating diffusion node **222**.

[0037] The source follower gate **224** functions as a high impedance amplifier for the pixel sensor **200**. The source follower gate **224** provides a voltage to current conversion of the floating diffusion voltage. The output of the source follower gate **224** is electrically connected with a row select gate **226**, which is configured to control the flow of the photocurrent **210** to external circuitry. The row select gate **226** is controlled by selectively applying a select voltage (V_{st}) **228** to the gate of the row select gate **226**. This permits the photocurrent **210** to flow to an output **230** of the pixel sensor **200**.

[0038] As indicated above, FIG. 2 is provided as an example. Other examples may differ from what is described with regard to FIG. 2.

[0039] FIGS. 3A and 3B are diagrams of an example implementation **300** of a pixel sensor array **302** described herein. FIG. 3A illustrates a top-down view of the pixel sensor array **302** and FIG. 3B illustrates a perspective view of the pixel sensor array **302**. In some implementations, the pixel sensor array **302** may be included in an image sensor **304**. The image sensor **304** may include a CMOS image sensor (CIS), a backside illuminated (BSI) CMOS image sensor, a front side illuminated (FSI) CMOS image sensor, or another type of image sensor.

[0040] As shown in FIGS. 3A and 3B, the pixel sensor array **302** may include a plurality of pixel sensors **200**. As further shown in FIGS. 3A and 3B, the pixel sensors **200** may be arranged in a grid. In some implementations, the pixel sensors **200** are square-shaped (as shown in the example in FIGS. 3A and 3B). In some implementations, the pixel sensors **200** include other shapes such as rectangle shapes, circle shapes, octagon shapes, diamond shapes, and/or other shapes.

[0041] The pixel sensors **200** may be configured to sense and/or accumulate incident light (e.g., light directed toward the pixel sensor array **302**). For example, a pixel sensor **200** may absorb and accumulate photons of the incident light in a photodiode **212**. The accumulation of photons in the photodiode **212** may generate a charge (e.g., a photocurrent) representing the intensity or brightness of the incident light (e.g., a greater amount of charge may correspond to a greater intensity or brightness, and a lower amount of charge may correspond to a lower intensity or brightness). The photocurrent may be transferred to a floating diffusion node **222** for storage, for amplification, and/or for transfer to another

node. In some implementations, at least a subset of the pixel sensors **200** may be configured to sense incident light in the visible light wavelength spectrum. In some implementations, at least a subset of the pixel sensors **200** may be configured to sense incident light in the infrared light or near infrared light wavelength spectrum.

[0042] In some implementations, the size of the pixel sensors **200** (e.g., the width or the diameter) of the pixel sensors **200** is less than approximately 1 micron. In these examples, the pixel sensors **200** may be referred to as sub-micron pixel sensors. Sub-micron pixel sensors may decrease the pixel sensor pitch (e.g., the distance between adjacent pixel sensors) in the pixel sensor array **302**, which may enable increased pixel sensor density in the pixel sensor array **302** (which can increase the performance of the pixel sensor array **302**).

[0043] As shown in FIGS. 3A and 3B, the pixel sensors **200** may be electrically and optically isolated by a BDTI structure **306** included in the pixel sensor array **302**. The BDTI structure **306** may include a plurality of interconnected and intersecting trenches that are filled with one or more types of materials, such as a dielectric material (e.g., an oxide-containing material, a high dielectric constant (high-k) dielectric material), a polysilicon material, and/or another type of material. The trenches of the BDTI structure **306** may be included around the perimeters of the pixel sensors **200** such that the BDTI structure **306** surrounds the photodiodes **212** of the pixel sensors **200**, as shown in FIG. 3A.

[0044] As further shown in FIGS. 3A and 3B, portions of the BDTI structure **306** extend along a side of one or two pixel sensors **200**. These portions of the BDTI structure **306** are referred to as non-cross-road portions (or non-X-road portions) **308** of the BDTI structure **306**. Portions of the BDTI structure **306** that intersect at corners of three or more pixel sensors **200** are referred to as cross-road portions (or X-road portions) **310** of the BDTI structure **306**.

[0045] As shown in FIG. 3A, a floating diffusion node **222** of a pixel sensor **200** may be included between corners of a plurality of pixel sensors **200** in a cross-road portion of the BDTI structure **306**. In particular, the floating diffusion node **222** is included under the cross-road portion of the BDTI structure **306**. Including the floating diffusion node **222** under the BDTI structure **306** (e.g., in the cross-road portion **310**) provides more area within the perimeter of the BDTI structure **306** for the photodiode **212** of the pixel sensor **200**, which enables the photodiode **212** to be larger than if the floating diffusion node **222** were also included within the perimeter of the BDTI structure **306**. This provides greater quantum efficiency and greater light sensitivity for the pixel sensor **200**. Alternatively, for the same size photodiode **212**, the area of the pixel sensor **200** may be smaller, thereby enabling a greater density of pixel sensors **200** to be included in the pixel sensor array **302**.

[0046] The BDTI structure **306** may extend into (and may be included in) a substrate (not shown) in which the pixel sensors **200** are formed to surround the photodiodes **212** and other structures of the pixel sensors **200** in the substrate. As indicated above, the pixel sensor array **302** may be included in a BSI CMOS image sensor. In these examples, the BDTI structure **306** may be formed from the backside of substrate of the pixel sensor array **302**.

[0047] FIGS. 3A and 3B further illustrate a reference cross-section A-A and a reference cross-section B-B that are

used in one or more figures described herein. Cross-section A-A is in a plane across a plurality of pixel sensors **200** of the pixel sensor array **302**. The plane extends across one or more non-cross-road portions **308** and one or more cross-road portions **310** of the BDTI structure **306**. Subsequent figures refer to this reference cross-section for clarity. Cross-section B-B is in a plane across the BDTI structure **306**, and includes one or more non-cross-road portions **308** and one or more cross-road portions **310** of the BDTI structure **306**. The cross-section B-B is used in subsequent figures to illustrate an elevation view in which a cross-section C-C is superimposed in background with the cross-section B-B in the foreground. In some figures, some reference numbers of components or features illustrated therein may be omitted to avoid obscuring other components or features for ease of depicting the figures.

[0048] As indicated above, FIGS. 3A and 3B are provided as an example. Other examples may differ from what is described with regard to FIGS. 3A and 3B.

[0049] FIGS. 4A and 4B are diagrams of an example implementation **400** of a portion of the pixel sensor array **302** described herein. FIG. 4A illustrates a cross-sectional view of the cross-section A-A of the portion of the pixel sensor array **302**. FIG. 4B illustrates an elevation view of the portion of the pixel sensor array **302** in which the cross-section C-C is superimposed in background the figures with the cross-section B-B in the foreground.

[0050] As shown in FIG. 4A, the pixel sensors **200** of the pixel sensor array **302** may be formed in and/or on a semiconductor layer **402**, which may be the substrate of the pixel sensor array **302**. The semiconductor layer **402** may include a silicon (Si) substrate, a substrate formed of a material including silicon, a III-V compound semiconductor material substrate such as gallium arsenide (GaAs), a germanium (Ge) substrate, a silicon germanium (SiGe) substrate, or another type of semiconductor substrate.

[0051] The pixel sensors **200** may each include a photodiode **212** in the semiconductor layer **402**. The photodiodes **212** between sections of the BDTI structure **306** such that the photodiodes **212** are optically isolated from one another. The photodiodes **212** may each include one or more of regions of the semiconductor layer **402** that are doped with various types of ions to form a p-n junction or a PIN junction (e.g., a junction between a p-type portion, an intrinsic (or undoped) type portion, and an n-type portion). For example, the semiconductor layer **402** may be doped with an n-type dopant to form one or more n-type regions of a photodiode **212**, and the semiconductor layer **402** may be doped with a p-type dopant to form a p-type region of the photodiode **212**. A photodiode **212** may be configured to absorb photons of incident light (e.g., visible light, near infrared light). The absorption of photons causes the photodiode **212** to accumulate a charge (referred to as a photocurrent **210**) due to the photoelectric effect. Photons may bombard the photodiode **212**, which causes emission of electrons and holes in the photodiode **212**, which causes the photocurrent **210** to be generated.

[0052] A floating diffusion node **222** may be included in a cross-road portion **310** at the corners of a plurality of pixel sensors **200**. In particular, the floating diffusion node **222** may be included under the cross-road portion **310** of the BDTI structure **306** between the BDTI structure **306** and a frontside surface of the semiconductor layer **402**. The floating diffusion node **222** may include a highly-doped n-type

region (e.g., an n⁺ doped region) of the semiconductor layer **402**. In some implementations, a drain extension region is included in the semiconductor layer **402** adjacent to the floating diffusion node **222**. The drain extension region may include lightly-doped n-type region(s) that facilitate the transfer of photocurrent **210** from the photodiode **212** to the floating diffusion node **222**.

[0053] A pixel sensor **200** may include a transfer gate **214** on the frontside surface of the semiconductor layer **402**. The transfer gate **214** may be configured to selectively control the transfer of the photocurrent **210** from a photodiode **212** of the pixel sensor **200** to the floating diffusion node **222** by selectively controlling the conductivity of the semiconductor layer **402** between the photodiode **212** and the floating diffusion node **222**. The transfer gates **214** of the pixel sensors **200** may be included in a dielectric layer **404** that is included on the frontside surface of the semiconductor layer **402**. The dielectric layer **404** may include a dielectric material such as a silicon oxide (SiO_x) (e.g., silicon dioxide (SiO₂)), a silicon nitride (Si₃N₄), a silicon carbide (SiC_x), a hafnium oxide (HfO_x), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), carbon doped silicon oxide, and/or another dielectric material.

[0054] A transfer gate **214** may include a gate electrode **406** and a gate dielectric layer **408** between the gate electrode **406** and the semiconductor layer **402**. In some implementations, the gate electrode **406** includes polysilicon. In some implementations, the gate electrode **406** includes one or more metals, such as tungsten (W), cobalt (Co), ruthenium (Ru), titanium (Ti), aluminum (Al), copper (Cu), gold (Au), aluminum (Al), an alloy thereof, among other examples of metals. The gate dielectric layer **408** may include a silicon nitride (Si₃N₄), a hafnium oxide (HfO_x), another high-k dielectric material, a silicon oxide (SiO_x), and/or another dielectric material.

[0055] Sidewall spacers **410** may be included on and/or around sidewalls of the gate electrodes **406**. The sidewall spacers **410** may include one or more dielectric materials such as a silicon oxide (SiO_x) (e.g., silicon dioxide (SiO₂)), a silicon nitride (Si₃N₄), a silicon carbide (SiC_x), a hafnium oxide (HfO_x), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), carbon doped silicon oxide, and/or another dielectric material.

[0056] As further shown in FIG. 4A, the BDTI structure **306** may extend into the semiconductor layer **402**. In particular, the BDTI structure **306** may extend into the semiconductor layer **402** from a backside surface of the semiconductor layer **402** and toward the frontside surface of the semiconductor layer **402**. The BDTI structure **306** is referred to as a backside DTI structure in that the BDTI structure **306** is formed into the semiconductor layer **402** from the backside surface of the semiconductor layer **402**. The BDTI structure **306** may be formed from the backside of the semiconductor layer **402** as part of backside processing of the pixel sensor array **302**. The backside of the semiconductor layer **402** may be the side of the semiconductor layer **402** opposing the front side on which the transfer gates **214** are included.

[0057] The BDTI structure **306** may include elongated structures of dielectric material **412** and a dielectric liner **414** between the dielectric material **412** and the semiconductor

layer **402**. The dielectric liner **414** may be included on sidewalls and on a bottom surface of the BDTI structure **306**, and may be included as an antireflective coating (ARC) and/or to further facilitate electrical and/or optical isolation of the pixel sensors **200**. In some implementations, the dielectric material **412** includes a silicon oxide (SiOx) (e.g., silicon dioxide (SiO₂)), a silicon nitride (Si_xN_y), a silicon carbide (SiC_x), a hafnium oxide (HfO_x), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), carbon doped silicon oxide, and/or another dielectric material. In some implementations, the dielectric liner **414** may include a high-k dielectric material such as a silicon nitride (Si_xN_y), a hafnium oxide (HfO_x), and/or another high-k dielectric material.

[0058] As further shown in FIG. 4A, dielectric etch stop structures **416** are included under the cross-road portions **310** of the BDTI structure **306** such that a dielectric etch stop structure **416** is located between a floating diffusion node **222** and a cross-road portion **310** of the BDTI structure **306**. A dielectric etch stop structure **416** is embedded in the semiconductor layer **402** between a floating diffusion node **222** and a cross-road portion **310** of the BDTI structure **306** to prevent or reduce the likelihood of etching into the floating diffusion node **222** when etching the semiconductor layer **402** to form trenches in which the BDTI structure **306** is to be formed. In particular, the dielectric etch stop structure **416** is included between a floating diffusion node **222** and a cross-road portion **310** of the BDTI structure **306** so that the etching of the trenches, in which the BDTI structure **306** is to be formed, stops on the dielectric etch stop structure **416** in the cross-road portion **310**, as opposed to etching into the underlying floating diffusion node **222** due to etch depth loading. In some implementations, the etching of the trenches, in which the BDTI structure **306** is to be formed, consumes a portion of the dielectric etch stop structure **416** (e.g., without the etching punching through the dielectric etch stop structure **416**). In these implementations, the bottom of the BDTI structure **306** in the cross-road portion **310** may extend into a portion of the dielectric etch stop structure **416**.

[0059] The dielectric etch stop structures **416** include one or more dielectric materials, such as a silicon oxide (SiOx) (e.g., silicon dioxide (SiO₂)), a silicon nitride (Si_xN_y), a silicon carbide (SiC_x), a hafnium oxide (HfO_x), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), carbon doped silicon oxide, and/or another dielectric material. The material(s) of the dielectric etch stop structures **416** are selected to have etch selectivity, relative to the semiconductor layer **402**, for the etchants that are used to etch the semiconductor layer **402** to form the trenches. For example, an etchant that has a high etch rate for silicon (Si) may be used to etch the semiconductor layer **402** to form the trenches. Thus, the material(s) of the dielectric etch stop structures **416** may be selected such that the etchants that are used to etch the semiconductor layer **402** have a low etch rate for the dielectric etch stop structures **416**.

[0060] As further shown in FIG. 4A, a dielectric etch stop structure **416** may have a dimension D1 corresponding to a cross-sectional width of the dielectric etch stop structure **416**. The dielectric etch stop structure **416** is formed such that the cross-sectional width of the dielectric etch stop

structure **416** results in the cross-road portion **310** of the BDTI structure **306** being fully within a perimeter of the dielectric etch stop structure **416**. This ensures that the dielectric etch stop structure **416** is physically large enough to fully contain the etching of the trenches of the BDTI structure **306** in the cross-road portion **310** to minimize and/or prevent etching of the trenches of the BDTI structure **306** below the dielectric etch stop structure **416** and into the underlying floating diffusion node **222**. The dielectric etch stop structure **416** is formed such that the cross-sectional width of the dielectric etch stop structure **416** results in the floating diffusion node **222** being fully within a perimeter of the dielectric etch stop structure **416** for similar purposes.

[0061] In some implementations, the cross-sectional width of the dielectric etch stop structure **416** (e.g., the dimension D1) is included in a range of approximately 200 nanometers to approximately 250 nanometers. If the cross-sectional width is less than approximately 200 nanometers, the dielectric etch stop structure **416** may not provide sufficient etching protection to prevent etching into the floating diffusion node **222**. If the cross-sectional width is greater than approximately 250 nanometers, the dielectric etch stop structure **416** may block some of the photocurrent **210** from being transferred from the photodiodes **212** to the floating diffusion node **222**, which may reduce the efficiency and light sensitivity of the pixel sensors **200**. If the cross-sectional width is included in the range of approximately 200 nanometers to approximately 250 nanometers, the dielectric etch stop structure **416** may provide sufficient etching protection while enabling a high efficiency and light sensitivity to be achieved for the pixel sensors **200**. However, other values for the cross-sectional width, and ranges other than approximately 200 nanometers to approximately 250 nanometers, are within the scope of the present disclosure.

[0062] As further shown in FIG. 4A, a dielectric etch stop structure **416** may have another dimension D2 corresponding to a thickness of the dielectric etch stop structure **416**. In some implementations, the thickness of the dielectric etch stop structure **416** is included in a range of approximately 50 nanometers to approximately 200 nanometers. If the thickness is less than approximately 50 nanometers, the dielectric etch stop structure **416** may not provide sufficient etching protection to prevent etching into the floating diffusion node **222**. If the thickness is greater than approximately 200 nanometers, the dielectric etch stop structure **416** may block some of the photocurrent **210** from being transferred from the photodiodes **212** to the floating diffusion node **222**, which may reduce the efficiency and light sensitivity of the pixel sensors **200**. If the thickness is included in the range of approximately 50 nanometers to approximately 200 nanometers, the dielectric etch stop structure **416** may provide sufficient etching protection while enabling a high efficiency and light sensitivity to be achieved for the pixel sensors **200**. However, other values for the thickness, and ranges other than approximately 200 nanometers to approximately 250 nanometers, are within the scope of the present disclosure.

[0063] As further shown in FIG. 4A, a buffer layer **418** may be included over the backside of the semiconductor layer **402**. The buffer layer **418** may include a silicon oxide (SiOx) (e.g., silicon dioxide (SiO₂)), a silicon nitride (Si_xN_y), a silicon carbide (SiC_x), a hafnium oxide (HfO_x), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass

(BPSG), fluorinated silica glass (FSG), carbon doped silicon oxide, and/or another dielectric material.

[0064] A composite metal grid **420** is included above the buffer layer **418**. The composite metal grid **420** includes a grid of intersecting columns that substantially conforms to the shape of the grid of the BDTI structure **306**. The composite metal grid **420** may include a layer stack of a dielectric portion **424** and a metal portion **422**. Alternatively, a metal grid with only a metal portion **422** may be included above the buffer layer **418**. A liner **426** may be included over the composite metal grid **420**.

[0065] The metal portion **422** may include one or more metals, such as tungsten (W), cobalt (Co), ruthenium (Ru), titanium (Ti), aluminum (Al), copper (Cu), gold (Au), aluminum (Al), an alloy thereof, among other examples of metals. The dielectric portions **424** may include a silicon oxide (SiOx) (e.g., silicon dioxide (SiO₂)), a silicon nitride (Si_xN_y), a silicon carbide (SiC_x), a hafnium oxide (HfO_x), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), carbon doped silicon oxide, and/or another dielectric material. The liner **426** may include a silicon oxide (SiOx) (e.g., silicon dioxide (SiO₂)), a silicon nitride (Si_xN_y), a silicon carbide (SiC_x), a hafnium oxide (HfO_x), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), carbon doped silicon oxide, and/or another dielectric material.

[0066] As further shown in FIG. 4, color filters **428** may be recessed in the composite metal grid **420** and may be included between the columns of the composite metal grid **420**. In some implementations, the color filters **428** include visible light color filters configured to filter a particular wavelength or a particular wavelength range of visible light (e.g., red light, blue light, or green light). In some implementations, at least a subset of the color filters **428** includes a near infrared (NIR) filter (e.g., an NIR bandpass filter) configured to permit wavelengths associated with NIR light to pass through the color filters **428** and to block other wavelengths of light. In some implementations, at least a subset of the color filters **428** includes an NIR cut filter configured to block NIR light from passing through the color filters **428**. In some implementations, color filters **428** may be omitted from one or more pixel sensors **200** to permit all wavelengths of light to pass through to the associated photodiodes **212**. In these examples, the pixel sensor(s) **200** may be configured as a white pixel sensor(s).

[0067] Micro-lenses **430** may be included above and/or on the color filters **428**. The micro-lenses **430** may include micro-lenses for the pixel sensors **200** configured to focus incident light toward the photodiodes **212** and/or to reduce optical crosstalk between the pixel sensors **200**.

[0068] As shown in FIG. 4B, the depth to which the BDTI structure **306** extends in the semiconductor layer **402** may be greater in the cross-road portions **310** than in the non-cross-road portions **308**. However, the inclusion of the dielectric etch stop structures **416** under the cross-road portions **310** of the BDTI structure **306** reduces a difference in depth of the cross-road portions **310** and the non-cross-road portions **308** (corresponding to dimension D3 in FIG. 4B) than if no dielectric etch stop structures **416** were included. The reduced difference in depth results in minimal to no etching into the underlying floating diffusion nodes **222**.

[0069] As indicated above, FIGS. 4A and 4B are provided as an example. Other examples may differ from what is described with regard to FIGS. 4A and 4B.

[0070] FIGS. 5A-5N are diagrams of an example implementation **500** of forming the pixel sensor array **302** of the image sensor device **304** described herein. FIGS. 5A-5N illustrate cross-section views of the example implementation **500** along the cross-section A-A of the pixel sensor array **302** in FIG. 4A. In some implementations, one or more processing operations described in connection with FIGS. 5A-5N may be performed using one or more of the semiconductor processing tools **102-114** illustrated and described in connection with FIG. 1. In some implementations, one or more processing operations described in connection with FIGS. 5A-5N may be performed using another semiconductor processing tool.

[0071] Turning to FIG. 5A, the example implementation **500** of forming the pixel sensor array **302** may be performed in connection with the semiconductor layer **402**. The semiconductor layer **402** may be provided as a semiconductor wafer or another type of semiconductor work piece. One or more processing operations described in connection with FIGS. 5A-5N may be performed from a frontside surface **502** of the semiconductor layer **402**, and one or more other processing operations described in connection with FIGS. 5A-5N may be performed from a backside surface **504** of the semiconductor layer **402**. The backside surface **504** may be a surface of the semiconductor layer **402** opposing the frontside surface **502**.

[0072] As shown in FIG. 5B, a plurality of regions of the semiconductor layer **402** may be doped to form photodiodes **212** for one or more pixel sensors **200**. For example, an ion implantation tool **114** may be used to dope the semiconductor layer **402** to form one or more n-type regions and/or one or more p-type regions of the photodiodes **212**. The ion implantation tool **114** may be used to implant p⁺ ions in the semiconductor layer **402** through the frontside surface **502** to form the p-type region(s) and/or may be used to implant n⁺ ions in the semiconductor layer **402** to form the n-type region(s).

[0073] As shown in FIG. 5C, a masking layer **506** may be formed on the frontside surface **502** of the semiconductor layer **402**. A deposition tool **102** may be used to deposit the masking layer **506** using a PVD technique, an ALD technique, a CVD technique, a spin-coating technique, an oxidation technique, another type of deposition technique described in connection with FIG. 1, and/or another suitable deposition technique. The masking layer **506** may be deposited in one or more deposition operations. The masking layer **506** may include a photoresist, a hard mask, and/or another type of masking layer.

[0074] As further shown in FIG. 5C, a pattern may be formed in the masking layer **506**. The pattern may be formed by removing portions of the masking layer **506** to expose portions of the frontside surface **502** of the semiconductor layer **402**. An exposure tool **104** may be used to expose the masking layer **506** to a radiation source to pattern the masking layer **506**. A developer tool **106** may be used to develop and remove portions of the masking layer **506** to expose the pattern. Additionally and/or alternatively, the pattern may be formed in a photoresist layer, and the pattern may be transferred to the masking layer **506** by etching the masking layer **506** (e.g., using an etch tool **108**) based on the pattern in the photoresist layer.

[0075] As shown in FIG. 5D, one or more ion implantation operations may be performed to form a plurality of implant regions 508 in the semiconductor layer 402 from the frontside surface 502. An ion implantation operations may be performed to implant ions in the semiconductor layer 402 to form the implant regions 508 based on the pattern in the masking layer 506. Thus, the masking layer 506 functions as an implant mask in the one or more ion implantation operations. In some implementations, the ions include oxygen ions, and the oxygen ions are implanted to form an oxide of the semiconductor material of the semiconductor layer 402. The implantation of the oxygen ions oxygenates the material of the semiconductor layer 402 to form the implant regions 508. For example, the oxygen ions may form silicon dioxide (SiO_2) from the silicon (Si) material of the semiconductor layer 402. In some implementations, the ions include nitrogen ions, and the nitrogen ions are implanted to form a nitride of the semiconductor material of the semiconductor layer 402. The implantation of the nitrogen ions nitrogenizes the material of the semiconductor layer 402 to form the implant regions 508. For example, the nitrogen ions may form silicon nitride (Si_3N_4) from the silicon (Si) material of the semiconductor layer 402.

[0076] As shown in FIG. 5E, a thermal operation may be performed to anneal the implant regions 508 to form a dielectric etch stop structure 416 from the implant regions 508. The thermal operation may include increasing the temperature of the semiconductor layer 402 for a time duration to cause the material of the implant regions 508 to diffuse and merge into a singular structure corresponding to the dielectric etch stop structure 416.

[0077] As further shown in FIG. 5E, the dielectric etch stop structure 416 may be formed a distance (corresponding to a dimension D4) below the frontside surface 502 of the semiconductor layer 402 and a distance (corresponding to a dimension D5) from the backside surface 504 of the semiconductor layer 402. The distance below the frontside surface 502 (the dimension D4) may be included in a range of approximately 200 nanometers to approximately 1000 nanometers. If the distance below the frontside surface 502 is less than approximately 200 nanometers, insufficient space may be provided for forming a floating diffusion node 222 between the dielectric etch stop structure 416 and the frontside surface 502 of the semiconductor layer 402. If the distance below the frontside surface 502 is greater than approximately 1000 nanometers, insufficient space may be provided for forming a BDTI structure 306 between the dielectric etch stop structure 416 and the backside surface 504 of the semiconductor layer 402. If the distance below the frontside surface 502 is included in the range of approximately 200 nanometers to approximately 1000 nanometers, sufficient area may be provided in the semiconductor layer 402 for formation of the floating diffusion node 222 and the BDTI structure 306. However, other values for the distance below the frontside surface 502, and ranges other than approximately 200 nanometers to approximately 1000 nanometers, are within the scope of the present disclosure.

[0078] The distance above the backside surface 504 (the dimension D5) may be greater than approximately 2400 nanometers. If the distance above the backside surface 504 is less than approximately 2400 nanometers, insufficient space may be provided for forming a BDTI structure 306 between the dielectric etch stop structure 416 and the backside surface 504 of the semiconductor layer 402. How-

ever, other values for the distance above the backside surface 504 are within the scope of the present disclosure.

[0079] As shown in FIG. 5F, one or more regions of the semiconductor layer 402 above the dielectric etch stop structures 416 may be doped to form floating diffusion nodes 222 of the one or more pixel sensors 200. In some implementations, an ion implantation tool 114 may be used to dope by implanting n^+ ions in the semiconductor layer 402 to form the floating diffusion nodes 222. The floating diffusion nodes 222 may be formed from the frontside surface 502 of the semiconductor layer 402. A distance between the frontside surface 502 of the semiconductor layer 402 and a floating diffusion node 222 (corresponding to dimension D6 in FIG. 5F) is less than the distance between the frontside surface 502 of the semiconductor layer 402 and a dielectric etch stop structure 416 (corresponding to dimension D4 in FIG. 5F). A distance between the frontside surface 502 of the semiconductor layer 402 and a floating diffusion node 222 (corresponding to dimension D6 in FIG. 5F) may also less than the distance between the frontside surface 502 of the semiconductor layer 402 and a photodiode 212 (corresponding to dimension D7 in FIG. 5F).

[0080] As shown in FIG. 5G, gate dielectric layers 408 of transfer gates 214 of the pixel sensors 200 may be formed over and/or on the frontside surface 502 of the semiconductor layer 402. Moreover, gate electrodes 406 of the transfer gates 214 may be formed over and/or on the gate dielectric layers 408. A deposition tool 102 may be used to deposit the gate dielectric layers 408 in a PVD operation, an ALD operation, a CVD operation, an oxidation operation, or another type of deposition operation. In some implementations, a planarization tool 110 may be used to planarize the gate dielectric layers 408 after the gate dielectric layers 408 are deposited. A deposition tool 102 and/or a plating tool 112 may be used to deposit the gate electrodes 406 using a CVD technique, a PVD technique, an ALD technique, an electroplating technique, another deposition technique described above in connection with FIG. 1, and/or another suitable deposition technique.

[0081] As further shown in FIG. 5G, sidewall spacers 410 may be formed on sidewalls of the gate electrodes 406. A deposition tool 102 may be used to deposit the sidewall spacers 410 in a PVD operation, an ALD operation, a CVD operation, an oxidation operation, and/or another type of deposition operation. In some implementations, a conformal layer of dielectric material is deposited, and then an etch tool 108 is used to remove portions of the dielectric layer, where the remaining portions of the dielectric layer correspond to the sidewall spacers 410. Accordingly, the sidewall spacers 410 may have a rounded outer surface resulting from the etch operation.

[0082] As shown in FIG. 5H, a dielectric layer 404 may be deposited on the frontside surface 502 of the semiconductor layer 402 and over the transfer gates 214. A deposition tool 102 may be used to deposit the dielectric layer 404 using a PVD technique, an ALD technique, a CVD technique, an oxidation technique, another type of deposition technique described in connection with FIG. 1, and/or another suitable deposition technique. The dielectric layer 404 may be deposited in one or more deposition operations. In some implementations, a planarization tool 110 may be used to planarize the dielectric layer 404 after the dielectric layer 404 is deposited.

[0083] FIGS. 5A-5H may correspond to frontside processing of the pixel sensor array 302 of the image sensor device 304. FIGS. 5I-5N may correspond to backside processing of the pixel sensor array 302 of the image sensor device 304.

[0084] As shown in FIG. 5I, a masking layer stack 510 is formed over and/or on the backside surface 504 of the semiconductor layer 402. The masking layer stack 510 may be used to form the trenches in which the BDTI structure 306 is to be formed. The masking layer stack 510 may include an antireflection layer 512, a bottom layer 514, a middle layer 516, and/or a photoresist layer 518, among other examples. The masking layer stack 510 may include the layers 512-518 to enable the trenches of the BDTI structure 306 to be formed to have a high or very high aspect ratio of a depth of the trenches to a width of the trenches.

[0085] A deposition tool 102 may be used to deposit the antireflection layer 512, a bottom layer 514, a middle layer 516, and/or a photoresist layer 518 using a PVD technique, an ALD technique, a CVD technique, a spin-coating technique, an oxidation technique, another type of deposition technique described in connection with FIG. 1, and/or another suitable deposition technique. The antireflection layer 512, a bottom layer 514, a middle layer 516, and/or a photoresist layer 518 may be deposited in one or more deposition operations. In some implementations, a planarization tool 110 may be used to planarize the antireflection layer 512, a bottom layer 514, a middle layer 516, and/or a photoresist layer 518 after the antireflection layer 512, a bottom layer 514, a middle layer 516, and/or a photoresist layer 518 are deposited.

[0086] As further shown in FIG. 5I, a pattern may be formed in the photoresist layer 518. The pattern may be formed by removing portions of the photoresist layer 518. An exposure tool 104 may be used to expose the photoresist layer 518 to a radiation source to pattern the photoresist layer 518. A developer tool 106 may be used to develop and remove portions of the photoresist layer 518 to expose the pattern. The pattern may be subsequently transferred to the antireflection layer 512, a bottom layer 514, and/or a middle layer 516.

[0087] As shown in FIG. 5J, an etch tool 108 may be used to etch the semiconductor layer 402 from the backside surface 504 of the semiconductor layer 402 to form trenches 520 and 522 in the semiconductor layer 402. The trenches 520 are formed in non-cross-road portions 308, and the trenches 522 are formed in cross-road portions 310. In the cross-road portions 310, etching of the trenches 522 stops on the dielectric etch stop structures 416. Thus, the dielectric etch stop structures 416 minimize and/or prevent the etching of the trenches 522 from extending into the underlying floating diffusion nodes 222. In some implementations, the etching of the trenches 522 extends into a portion of the dielectric etch stop structures 416.

[0088] The trenches 520 and 522 are formed based on the pattern formed in the masking layer stack 510. In some implementations, an etch technique is used to form the trenches 520 and 522 to have a high or very high aspect ratio of a depth of the trenches to a width of the trenches 520 and 522. For example, a deep reactive ion etch technique (sometimes referred to as a "Bosch" etch technique) may be used to form the trenches 520 and 522. This technique may involve the use of sequential operations to incrementally increase the depth of the trenches 520 and 522 in a highly controlled manner. The sequential operations may include a

plurality of cycles of forming a protective film on the bottom surface and the sidewalls in the trenches 520 and 522, removing the protective film from the bottom surface, and increasing the depth of the trenches 520 and 522 as the protective film protects the sidewalls from being etched. This enables the depth of the trenches 520 and 522 to be increased with minimal increase to the width of the trenches 520 and 522. However, another etch technique may be used to form the trenches 520 and 522.

[0089] As shown in FIG. 5K, a dielectric liner 414 may be formed on sidewalls and on bottom surfaces of the trenches 520 and 522. In some implementations, portions of the dielectric liner 414 are also formed on the backside surface 504 of the semiconductor layer 402. A deposition tool 102 may be used to conformally deposit the dielectric liner 414 in a PVD operation, an ALD operation, a CVD operation, and/or another type of deposition operation. In some implementations, the dielectric liner 414 is subsequently removed from the backside surface 504 of the semiconductor layer 402. In some implementations, the dielectric liner 414 remains on the backside surface 504 of the semiconductor layer 402 (e.g., as an antireflective coating).

[0090] As shown in FIG. 5L, the trenches 520 and 522 may be filled with a dielectric material 412 (e.g., an oxide-containing dielectric material, a high-k dielectric material) over the dielectric liner 414 to form the BDTI structure 306 in the trenches 520 and 522. The BDTI structure 306 may be formed on the dielectric etch stop structures 416 in the cross-road portions 310. As further shown in FIG. 5L, the dielectric material 412 may be deposited over the backside surface 504 of the semiconductor layer 402 to form the buffer layer 418. A deposition tool 102 may be used to deposit the dielectric material 412 in the trenches 520 and 522 to form the BDTI structure 306 in a PVD operation, an ALD operation, a CVD operation, an oxidation operation, and/or another type of deposition operation. A deposition tool 102 may be used to deposit the buffer layer 418 in a PVD operation, an ALD operation, a CVD operation, an oxidation operation, and/or another type of deposition operation. In some implementations, a planarization tool 110 may be used to planarize the buffer layer 418 after the buffer layer 418 is deposited.

[0091] As shown in FIG. 5M, a composite metal grid 420 may be formed above the buffer layer 418. Forming the composite metal grid 420 may include depositing the layers of the composite metal grid 420 and using a pattern in a masking layer to etch the layers to form the metal portion 422 and/or the dielectric portion 424 of the composite metal grid 420. The liner 426 may then be conformally deposited on the metal portion 422 and/or on the dielectric portion 424.

[0092] As shown in FIG. 5N, color filters 428 may be formed in the composite metal grid 420 above the pixel sensors 200, and micro-lenses 430 may be formed over and/or on the color filters 428.

[0093] As indicated above, FIGS. 5A-5N are provided as an example. Other examples may differ from what is described with regard to FIGS. 5A-5N.

[0094] FIGS. 6A-6G are diagrams of an example implementation 600 of forming the pixel sensor array 302 of the image sensor device 304 described herein. FIGS. 6A-6G illustrate cross-section views of the example implementation 600 along the cross-section A-A of the pixel sensor array 302 in FIG. 4A. In some implementations, one or more processing operations described in connection with FIGS. 6A-6G

may be performed using one or more of the semiconductor processing tools **102-114** illustrated and described in connection with FIG. 1. In some implementations, one or more processing operations described in connection with FIGS. 6A-6G may be performed using another semiconductor processing tool.

[0095] As shown in FIG. 6A, a plurality of regions of the semiconductor layer **402** may be doped to form photodiodes **212** for one or more pixel sensors **200**. For example, an ion implantation tool **114** may be used to dope the semiconductor layer **402** to form one or more n-type regions and/or one or more p-type regions of the photodiodes **212**. The ion implantation tool **114** may be used to implant p⁺ ions in the semiconductor layer **402** through the frontside surface **502** to form the p-type region(s) and/or may be used to implant n⁺ ions in the semiconductor layer **402** to form the n-type region(s).

[0096] As further shown in FIG. 6A, a masking layer **602** may be formed on the frontside surface **502** of the semiconductor layer **402**. A deposition tool **102** may be used to deposit the masking layer **602** using a PVD technique, an ALD technique, a CVD technique, a spin-coating technique, an oxidation technique, another type of deposition technique described in connection with FIG. 1, and/or another suitable deposition technique. The masking layer **602** may be deposited in one or more deposition operations. The masking layer **602** may include a photoresist, a hard mask, and/or another type of masking layer.

[0097] As further shown in FIG. 6A, a pattern may be formed in the masking layer **602**. The pattern may be formed by removing portions of the masking layer **602** to expose portions of the frontside surface **502** of the semiconductor layer **402**. An exposure tool **104** may be used to expose the masking layer **506** to a radiation source to pattern the masking layer **602**. A developer tool **106** may be used to develop and remove portions of the masking layer **602** to expose the pattern. Additionally and/or alternatively, the pattern may be formed in a photoresist layer, and the pattern may be transferred to the masking layer **602** by etching the masking layer **602** (e.g., using an etch tool **108**) based on the pattern in the photoresist layer.

[0098] As shown in FIG. 6B, the semiconductor layer **402** may be etched from the frontside surface **502** to form a recess **604** in the semiconductor layer **402**. The semiconductor layer **402** may be etched (e.g., using an etch tool **108**) to form the recess **604** based on the pattern in the masking layer **602**. The semiconductor layer **402** may be etched using a wet etch technique, a dry etch technique, and/or another etch technique.

[0099] As shown in FIG. 6C, the recess **604** is filled with the material of a dielectric etch stop structure **416**. In some implementations, the recess **604** is fully filled with the material of the dielectric etch stop structure **416** such that the dielectric etch stop structure **416** extends above the frontside surface **502** of the semiconductor layer **402**. This reduces the likelihood of void formation in the dielectric etch stop structure **416**. A deposition tool **102** may be used to deposit the material of the dielectric etch stop structure **416** using a PVD technique, an ALD technique, a CVD technique, an oxidation technique, another type of deposition technique described in connection with FIG. 1, and/or another suitable deposition technique. The material of the dielectric etch stop structure **416** may be deposited in one or more deposition operations.

[0100] As shown in FIG. 6D, an etch-back operation may be performed to remove material from the dielectric etch stop structure **416**. This results in the dielectric etch stop structure **416** occupying less than an entirety of the area in the recess **604**. The etch-back operation may be performed to reduce the thickness of the dielectric etch stop structure **416** (e.g., to a target thickness or so that the thickness of the dielectric etch stop structure **416** is included in a particular range described herein) and/or to provide sufficient area above the dielectric etch stop structure **416** for formation of a floating diffusion node **222** between the dielectric etch stop structure **416** and the frontside surface **502** of the semiconductor layer **402**. The dielectric etch stop structure **416** may be etched using an etch tool **108**. The dielectric etch stop structure **416** may be etched using a wet etch technique, a dry etch technique, and/or another etch technique.

[0101] As shown in FIG. 6E, a portion **606** of the semiconductor layer **402** may be epitaxially regrown (e.g., using a deposition tool **102**) above the dielectric etch stop structure **416** in the recess **604**. In some implementations, the sidewalls of the recess **604** (corresponding to the semiconductor layer **402**) function as a substrate on which the material of the portion **606** is grown. The portion **606** may regrow laterally from the sidewalls in the recess **604** until the portion **606** coalesces into a continuous region of the semiconductor layer **402**.

[0102] As shown in FIG. 6F, a floating diffusion node **222** may be formed above the dielectric etch stop structure **416** in the portion **606** of the semiconductor layer **402** that was regrown above the dielectric etch stop structure **416**.

[0103] As shown in FIG. 6G, the transfer gates **214** (including the gate dielectric layers **408**, the gate electrodes **406**, and the sidewall spacers **410**), the BDTI structure **306** (including the dielectric material **412** and the dielectric liner **414**), the buffer layer **418**, the composite metal grid **420** (including the metal portion **422**, the dielectric portion **424**, and/or the liner **426**), the color filters **428**, and the microlenses **430** may be formed. Similar processing operations as illustrated and described in connection with FIGS. 5G-5N may be performed to form these structures.

[0104] As indicated above, FIGS. 6A-6G are provided as an example. Other examples may differ from what is described with regard to FIGS. 6A-6G.

[0105] FIG. 7 is a diagram of example components of a device **700** described herein. In some implementations, one or more of the semiconductor processing tools **102-114** and/or the wafer/die transport tool **116** may include one or more devices **700** and/or one or more components of the device **700**. As shown in FIG. 7, the device **700** may include a bus **710**, a processor **720**, a memory **730**, an input component **740**, an output component **750**, and/or a communication component **760**.

[0106] The bus **710** may include one or more components that enable wired and/or wireless communication among the components of the device **700**. The bus **710** may couple together two or more components of FIG. 7, such as via operative coupling, communicative coupling, electronic coupling, and/or electric coupling. For example, the bus **710** may include an electrical connection (e.g., a wire, a trace, and/or a lead) and/or a wireless bus. The processor **720** may include a central processing unit, a graphics processing unit, a microprocessor, a controller, a microcontroller, a digital signal processor, a field-programmable gate array, an application-specific integrated circuit, and/or another type of

processing component. The processor 720 may be implemented in hardware, firmware, or a combination of hardware and software. In some implementations, the processor 720 may include one or more processors capable of being programmed to perform one or more operations or processes described elsewhere herein.

[0107] The memory 730 may include volatile and/or non-volatile memory. For example, the memory 730 may include random access memory (RAM), read only memory (ROM), a hard disk drive, and/or another type of memory (e.g., a flash memory, a magnetic memory, and/or an optical memory). The memory 730 may include internal memory (e.g., RAM, ROM, or a hard disk drive) and/or removable memory (e.g., removable via a universal serial bus connection). The memory 730 may be a non-transitory computer-readable medium. The memory 730 may store information, one or more instructions, and/or software (e.g., one or more software applications) related to the operation of the device 700. In some implementations, the memory 730 may include one or more memories that are coupled (e.g., communicatively coupled) to one or more processors (e.g., processor 720), such as via the bus 710. Communicative coupling between a processor 720 and a memory 730 may enable the processor 720 to read and/or process information stored in the memory 730 and/or to store information in the memory 730.

[0108] The input component 740 may enable the device 700 to receive input, such as user input and/or sensed input. For example, the input component 740 may include a touch screen, a keyboard, a keypad, a mouse, a button, a microphone, a switch, a sensor, a global positioning system sensor, a global navigation satellite system sensor, an accelerometer, a gyroscope, and/or an actuator. The output component 750 may enable the device 700 to provide output, such as via a display, a speaker, and/or a light-emitting diode. The communication component 760 may enable the device 700 to communicate with other devices via a wired connection and/or a wireless connection. For example, the communication component 760 may include a receiver, a transmitter, a transceiver, a modem, a network interface card, and/or an antenna.

[0109] The device 700 may perform one or more operations or processes described herein. For example, a non-transitory computer-readable medium (e.g., memory 730) may store a set of instructions (e.g., one or more instructions or code) for execution by the processor 720. The processor 720 may execute the set of instructions to perform one or more operations or processes described herein. In some implementations, execution of the set of instructions, by one or more processors 720, causes the one or more processors 720 and/or the device 700 to perform one or more operations or processes described herein. In some implementations, hardware circuitry may be used instead of or in combination with the instructions to perform one or more operations or processes described herein. Additionally, or alternatively, the processor 720 may be configured to perform one or more operations or processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

[0110] The number and arrangement of components shown in FIG. 7 are provided as an example. The device 700 may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. 7. Additionally, or alternatively, a

set of components (e.g., one or more components) of the device 700 may perform one or more functions described as being performed by another set of components of the device 700.

[0111] FIG. 8 is a flowchart of an example process 800 associated with forming an image sensor device described herein. In some implementations, one or more process blocks of FIG. 8 are performed using one or more semiconductor processing tools (e.g., one or more of the semiconductor processing tools 102-114). Additionally, or alternatively, one or more process blocks of FIG. 8 may be performed using one or more components of device 700, such as processor 720, memory 730, input component 740, output component 750, and/or communication component 760.

[0112] As shown in FIG. 8, process 800 may include forming, from a first surface of a semiconductor layer of an image sensor device, a dielectric etch stop structure in the semiconductor layer (block 810). For example, one or more of the semiconductor processing tools 102-114 may be used to form, from a first surface (e.g., a frontside surface 502) of a semiconductor layer 402 of an image sensor device 304, a dielectric etch stop structure 416 in the semiconductor layer 402, as described herein.

[0113] As further shown in FIG. 8, process 800 may include forming, from the first surface of the semiconductor layer, a floating diffusion node above the dielectric etch stop structure in the semiconductor layer (block 820). For example, one or more of the semiconductor processing tools 102-114 may be used to form, from the first surface of the semiconductor layer 402, a floating diffusion node 222 above the dielectric etch stop structure 416 in the semiconductor layer 402, as described herein.

[0114] As further shown in FIG. 8, process 800 may include etching the semiconductor layer, from a second surface (e.g., a backside surface 504) of the semiconductor layer opposing the first surface, to form a plurality of intersecting trenches in the semiconductor layer after forming the floating diffusion node (block 830). For example, one or more of the semiconductor processing tools 102-114 may be used to etch the semiconductor layer 402, from a second surface of the semiconductor layer 402 opposing the first surface, to form a plurality of intersecting trenches (e.g., trenches 520, trenches 522) in the semiconductor layer 402 after forming the floating diffusion node 222, as described herein. In some implementations, two or more of the intersecting trenches intersect in a cross-road portion 310 above the dielectric etch stop structure 416 and above the floating diffusion node 222. In some implementations, the etching of the semiconductor layer 402 stops on the dielectric etch stop structure 416 in the cross-road portion 310.

[0115] Process 800 may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

[0116] In a first implementation, forming the dielectric etch stop structure 416 includes performing an ion implantation operation to form the dielectric etch stop structure 416 below the first surface of the semiconductor layer 402.

[0117] In a second implementation, alone or in combination with the first implementation, performing the ion implantation operation includes performing the ion implantation

tation operation to implant oxygen ions in the semiconductor layer **402** to form the dielectric etch stop structure **416**.

[0118] In a third implementation, alone or in combination with one or more of the first and second implementations, performing the ion implantation operation includes performing the ion implantation operation to form a plurality of implant regions **508** in the semiconductor layer **402**.

[0119] In a fourth implementation, alone or in combination with one or more of the first through third implementations, forming the dielectric etch stop structure **416** includes performing a thermal operation to anneal the plurality of implant regions **508** to form the dielectric etch stop structure **416** from the plurality of implant regions **508**.

[0120] In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, forming the dielectric etch stop structure **416** includes forming the dielectric etch stop structure **416** in a recess **604** in the first surface of the semiconductor layer **402**.

[0121] In a sixth implementation, alone or in combination with one or more of the first through fifth implementations, process **800** includes epitaxially regrowing a portion **606** of the semiconductor layer **402** on the dielectric etch stop structure **416** in the recess **604** after forming the dielectric etch stop structure **416**.

[0122] In a seventh implementation, alone or in combination with one or more of the first through sixth implementations, forming the floating diffusion node **222** includes forming at least a portion of the floating diffusion node **222** in the portion **606** of the semiconductor layer **402** that was epitaxially regrown above the dielectric etch stop structure **416**.

[0123] Although FIG. **8** shows example blocks of process **800**, in some implementations, process **800** includes additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **8**. Additionally, or alternatively, two or more of the blocks of process **800** may be performed in parallel.

[0124] FIG. **9** is a flowchart of an example process **900** associated with image sensor device and methods of formation. In some implementations, one or more process blocks of FIG. **9** are performed using one or more semiconductor processing tools (e.g., one or more of the semiconductor processing tools **102-114**). Additionally, or alternatively, one or more process blocks of FIG. **9** may be performed using one or more components of device **700**, such as processor **720**, memory **730**, input component **740**, output component **750**, and/or communication component **760**.

[0125] As shown in FIG. **9**, process **900** may include forming, from a first surface of a semiconductor layer of an image sensor device, a plurality of photodiodes in the semiconductor layer (block **910**). For example, one or more of the semiconductor processing tools **102-114** may be used to form, from a first surface (e.g., a frontside surface **502**) of a semiconductor layer **402** of an image sensor device **304**, a plurality of photodiodes **212** in the semiconductor layer **402**, as described herein.

[0126] As further shown in FIG. **9**, process **900** may include forming, from the first surface, a dielectric etch stop structure adjacent to the plurality of photodiodes in the semiconductor layer (block **920**). For example, one or more of the semiconductor processing tools **102-114** may be used to form, from the first surface, a dielectric etch stop structure **416** adjacent to the plurality of photodiodes **212** in the semiconductor layer **402**, as described herein. In some

implementations, a distance (e.g., the dimension **D4**) between the dielectric etch stop structure **416** and the first surface is less than a distance (e.g., the dimension **D7**) between the plurality of photodiodes **212** and the first surface.

[0127] As further shown in FIG. **9**, process **900** may include forming, from the first surface of the semiconductor layer, a floating diffusion node above the dielectric etch stop structure in the semiconductor layer (block **930**). For example, one or more of the semiconductor processing tools **102-114** may be used to form, from the first surface of the semiconductor layer **402**, a floating diffusion node **222** above the dielectric etch stop structure **416** in the semiconductor layer **402**, as described herein.

[0128] As further shown in FIG. **9**, process **900** may include etching the semiconductor layer, from a second surface of the semiconductor layer opposing the first surface, to form a plurality of intersecting trenches in the semiconductor layer after forming the floating diffusion node (block **940**). For example, one or more of the semiconductor processing tools **102-114** may be used to etch the semiconductor layer **402**, from a second surface (e.g., a backside surface **504**) of the semiconductor layer **402** opposing the first surface, to form a plurality of intersecting trenches (e.g., trenches **520**, trenches **522**) in the semiconductor layer **402** after forming the floating diffusion node **222**, as described herein. In some implementations, two or more of the intersecting trenches intersect in a cross-road portion **310** above the dielectric etch stop structure **416** and above the floating diffusion node **222**. In some implementations, the etching of the semiconductor layer **402** stops on the dielectric etch stop structure **416** in the cross-road portion **310**.

[0129] As further shown in FIG. **9**, process **900** may include forming a BDTI structure in the plurality of intersecting trenches (block **950**). For example, one or more of the semiconductor processing tools **102-114** may be used to form a BDTI structure **306** in the plurality of intersecting trenches, as described herein. In some implementations, a portion of the BDTI structure **306**, in the cross-road portion **310**, is formed on the dielectric etch stop structure **416**.

[0130] Process **900** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

[0131] In a first implementation, forming the dielectric etch stop structure **416** includes forming, from the first surface of the semiconductor layer **402**, a recess **604** in the semiconductor layer **402**, and depositing the dielectric etch stop structure **416** in the recess **604**.

[0132] In a second implementation, alone or in combination with the first implementation, depositing the dielectric etch stop structure **416** in the recess **604** includes fully filling the recess with the dielectric etch stop structure **416**.

[0133] In a third implementation, alone or in combination with one or more of the first and second implementations, forming the dielectric etch stop structure **416** includes performing an etch-back operation to remove a portion of the dielectric etch stop structure **416** such that the dielectric etch stop structure **416** occupies less than an entirety of the recess **604**.

[0134] In a fourth implementation, alone or in combination with one or more of the first through third implementations, process **900** includes epitaxially regrowing a portion

606 of the semiconductor layer 402 on the dielectric etch stop structure 416 in the recess 604 after forming the dielectric etch stop structure 416.

[0135] In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, forming the dielectric etch stop structure 416 includes performing an ion implantation operation to form the dielectric etch stop structure 416 below the first surface of the semiconductor layer 402.

[0136] Although FIG. 9 shows example blocks of process 900, in some implementations, process 900 includes additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 9. Additionally, or alternatively, two or more of the blocks of process 900 may be performed in parallel.

[0137] In this way, dielectric regions are formed in a semiconductor substrate of a semiconductor device. The dielectric regions may be formed in locations above which cross-road portions of a BDTI structure are to be formed for a pixel sensor array of the semiconductor device. The dielectric regions may function as etch stop structures in the cross-road portions when etching the semiconductor substrate to form the trenches in which the BDTI structure is to be formed. In particular, the dielectric regions minimize and/or prevent the increase in trench depth of the trenches in the cross-road portions relative to the trench depth of the trenches in the non-cross-road portions. In this way, the dielectric regions reduce and/or minimize trench depth loading in the cross-road portions of the trenches. Thus, the dielectric regions prevent, minimize, and/or reduce the likelihood of etching into floating diffusion nodes of pixel sensors of the pixel sensor array, which reduces the likelihood of degraded performance and/or failure of the pixel sensors.

[0138] As described in greater detail above, some implementations described herein provide a method. The method includes forming, from a first surface of a semiconductor layer of an image sensor device, a dielectric etch stop structure in the semiconductor layer. The method includes forming, from the first surface of the semiconductor layer, a floating diffusion node above the dielectric etch stop structure in the semiconductor layer. The method includes etching the semiconductor layer, from a second surface of the semiconductor layer opposing the first surface, to form a plurality of intersecting trenches in the semiconductor layer after forming the floating diffusion node, where two or more of the intersecting trenches intersect in a cross-road portion above the dielectric etch stop structure and the floating diffusion node, and where the etching of the semiconductor layer stops on the dielectric etch stop structure in the cross-road portion.

[0139] As described in greater detail above, some implementations described herein provide a method. The method includes forming, from a first surface of a semiconductor layer of an image sensor device, a plurality of photodiodes in the semiconductor layer. The method includes forming, from the first surface, a dielectric etch stop structure adjacent to the plurality of photodiodes in the semiconductor layer, where a distance between the dielectric etch stop structure and the first surface is less than a distance between the plurality of photodiodes and the first surface. The method includes forming, from the first surface of the semiconductor layer, a floating diffusion node above the dielectric etch stop structure in the semiconductor layer. The method includes

etching the semiconductor layer, from a second surface of the semiconductor layer opposing the first surface, to form a plurality of intersecting trenches in the semiconductor layer after forming the floating diffusion node, where two or more of the intersecting trenches intersect in a cross-road portion above the dielectric etch stop structure and the floating diffusion node, and where the etching of the semiconductor layer stops on the dielectric etch stop structure in the cross-road portion. The method includes forming a BDTI structure in the plurality of intersecting trenches, where a portion of the BDTI structure, in the cross-road portion, is formed on the dielectric etch stop structure.

[0140] As described in greater detail above, some implementations described herein provide an image sensor device. The image sensor device includes a plurality of pixel sensors. The image sensor device includes a BDTI structure at least laterally surrounding a photodiode of a pixel sensor of the plurality of pixel sensors, where the BDTI structure is included partially through a semiconductor layer of the image sensor device between a front side of the semiconductor layer and a backside of the semiconductor layer. The image sensor device includes a dielectric etch stop structure under a cross-road portion of the BDTI structure.

[0141] As used herein, “satisfying a threshold” may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, not equal to the threshold, or the like.

[0142] The terms “approximately” and “substantially” can indicate a value of a given quantity that varies within 5% of the value (e.g., +1%, +2%, +3%, +4%, +5% of the value). These values are merely examples and are not intended to be limiting. It is to be understood that the terms “approximately” and “substantially” can refer to a percentage of the values of a given quantity in light of this disclosure.

[0143] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method, comprising:

forming, from a first surface of a semiconductor layer of an image sensor device, a dielectric etch stop structure in the semiconductor layer;

forming, from the first surface of the semiconductor layer, a floating diffusion node above the dielectric etch stop structure in the semiconductor layer; and

etching the semiconductor layer, from a second surface of the semiconductor layer opposing the first surface, to form a plurality of intersecting trenches in the semiconductor layer after forming the floating diffusion node,

wherein two or more of the intersecting trenches intersect in a cross-road portion above the dielectric etch stop structure and the floating diffusion node, and

wherein the etching of the semiconductor layer stops on the dielectric etch stop structure in the cross-road portion.

2. The method of claim 1, wherein forming the dielectric etch stop structure comprises:

performing an ion implantation operation to form the dielectric etch stop structure below the first surface of the semiconductor layer.

3. The method of claim 2, wherein performing the ion implantation operation comprises:

performing the ion implantation operation to implant oxygen ions in the semiconductor layer to form the dielectric etch stop structure.

4. The method of claim 2, wherein performing the ion implantation operation comprises:

performing the ion implantation operation to form a plurality of implant regions in the semiconductor layer.

5. The method of claim 4, wherein forming the dielectric etch stop structure comprises:

performing a thermal operation to anneal the plurality of implant regions to form the dielectric etch stop structure from the plurality of implant regions.

6. The method of claim 1, wherein forming the dielectric etch stop structure comprises:

forming the dielectric etch stop structure in a recess in the first surface of the semiconductor layer.

7. The method of claim 6, further comprising:

epitaxially regrowing a portion of the semiconductor layer on the dielectric etch stop structure in the recess after forming the dielectric etch stop structure.

8. The method of claim 7, wherein forming the floating diffusion node comprises:

forming at least a portion of the floating diffusion node in the portion of the semiconductor layer that was epitaxially regrown above the dielectric etch stop structure.

9. A method, comprising:

forming, from a first surface of a semiconductor layer of an image sensor device, a plurality of photodiodes in the semiconductor layer;

forming, from the first surface, a dielectric etch stop structure adjacent to the plurality of photodiodes in the semiconductor layer,

wherein a distance between the dielectric etch stop structure and the first surface is less than a distance between the plurality of photodiodes and the first surface;

forming, from the first surface of the semiconductor layer, a floating diffusion node above the dielectric etch stop structure in the semiconductor layer;

etching the semiconductor layer, from a second surface of the semiconductor layer opposing the first surface, to form a plurality of intersecting trenches in the semiconductor layer after forming the floating diffusion node,

wherein two or more of the intersecting trenches intersect in a cross-road portion above the dielectric etch stop structure and the floating diffusion node, and

wherein the etching of the semiconductor layer stops on the dielectric etch stop structure in the cross-road portion; and

forming a backside deep trench isolation (BDTI) structure in the plurality of intersecting trenches,

wherein a portion of the BDTI structure, in the cross-road portion, is formed on the dielectric etch stop structure.

10. The method of claim 9, wherein forming the dielectric etch stop structure comprises:

forming, from the first surface of the semiconductor layer, a recess in the semiconductor layer; and

depositing the dielectric etch stop structure in the recess.

11. The method of claim 10, wherein depositing the dielectric etch stop structure in the recess comprises:

fully filling the recess with the dielectric etch stop structure.

12. The method of claim 11, wherein forming the dielectric etch stop structure comprises:

performing an etch-back operation to remove a portion of the dielectric etch stop structure such that the dielectric etch stop structure occupies less than an entirety of the recess.

13. The method of claim 10, further comprising:

epitaxially regrowing a portion of the semiconductor layer on the dielectric etch stop structure in the recess after forming the dielectric etch stop structure.

14. The method of claim 9, wherein forming the dielectric etch stop structure comprises:

performing an ion implantation operation to form the dielectric etch stop structure below the first surface of the semiconductor layer.

15. An image sensor device, comprising:

a plurality of pixel sensors;

a backside deep trench isolation (BDTI) structure at least laterally surrounding a photodiode of a pixel sensor of the plurality of pixel sensors,

wherein the BDTI structure is included partially through a semiconductor layer of the image sensor device between a front side of the semiconductor layer and a backside of the semiconductor layer; and

a dielectric etch stop structure under a cross-road portion of the BDTI structure.

16. The image sensor device of claim 15, wherein the dielectric etch stop structure is between the cross-road portion of the BDTI structure and the front side of the semiconductor layer.

17. The image sensor device of claim 15, wherein a width of the dielectric etch stop structure is included in a range of approximately 200 nanometers to approximately 250 nanometers.

18. The image sensor device of claim 15, further comprising:

a floating diffusion node of the pixel sensor,

wherein the dielectric etch stop structure is between the cross-road portion of the BDTI structure and the floating diffusion node.

19. The image sensor device of claim 15, wherein a thickness of the dielectric etch stop structure is included in a range of approximately 50 nanometers to approximately 200 nanometers.

20. The image sensor device of claim 15, wherein the dielectric etch stop structure comprises an oxide of a semiconductor material of the semiconductor layer.

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